

Applications of the Strong Dominability Criterion and the Oka Union Theorem

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Abstract

The strong dominability criterion and the Oka union theorem yield new permanence mechanisms for complex manifolds. We prove descent through regular preimages, an Oka theorem for complements of product analytic subsets, and an equivalence for holomorphic families of connected complex Lie groups. These tools give Oka properties for moduli of stable and parabolic bundles, spaces of maps and embeddings, Hilbert and Quot schemes, rank-two representation and connection moduli, elliptic and abelian fibrations, compactified Jacobians for one-node integral families, and the Gross–Popescu derived pair. The proofs construct dominating entire maps or finite closed filtrations with Oka strata.

Contents

Introduction	2
1 Oka-theoretic mechanisms	2
1.1 Background and descent	2
1.2 Complements of product analytic subsets	4
1.3 Families and deformation spaces	6
2 Stable and parabolic bundles on curves	8
3 Holomorphic mapping and embedding spaces	16
3.1 Maps defined by sections	16
3.2 Embeddings and maps to Grassmannians	21
4 Hilbert and Quot schemes	26
5 Character varieties and parabolic connections	30
5.1 Character and representation varieties	30
5.2 Parabolic connections and Garnier spaces	34
6 Elliptic and abelian fibrations	38
6.1 Elliptic surfaces	38
6.2 Abelian fibrations	42
6.3 The Gross–Popescu Calabi–Yau pair	46
7 Compactified Jacobians	47

Introduction

The companion manuscript of Du, Guo, Wang, and Xie proves that a connected complex manifold is Oka exactly when it is strongly dominable [17, Theorem 1.3]. It also proves that the Oka property extends across a proper closed complex analytic subset of a connected complex manifold [17, Theorem 1.5]. Together with the finite-stratification consequence [17, Corollary 4.5], these results turn pointwise entire-map constructions and geometric decompositions into Oka theorems.

Oka's solution of the second Cousin problem isolated the topological obstruction behind the classical Oka principle [45]. Grauert's bundle theorem [27], Gromov's elliptic-bundle principle [29], and Forstnerič's convex-approximation characterization [19] are milestones in its development into modern Oka theory. The results of the companion manuscript supply the final step that converts the constructions in this note into Oka statements.

This note develops applications for which that final passage is essential. The first section establishes the common mechanisms. A regular value of a holomorphic map from an Oka manifold supplies a dominating entire map on the target. A triangular construction treats complements of product analytic subsets. Relative exponential maps compare a holomorphic family of connected complex Lie groups with its base.

The moduli applications begin with stable bundles on smooth curves and parabolic bundles. Explicit affine extension families cover the stable loci, while incidence estimates place the unstable sets in codimension at least two. Entire families of generating sections then give Oka spaces of maps to projective space, spaces of embeddings, and stable-quotient maps to Grassmannians. Intrinsic Hilbert–Samuel strata and an upper-semicontinuous profile produce a finite closed filtration of surface Hilbert schemes; an exact-rank filtration treats nested Quot schemes on curves.

For rank-two representation spaces, complete Goldman flows handle the irreducible character locus, and the reducible Schur locus is added as a closed Oka submanifold. The Riemann–Hilbert correspondence yields the analogous result for nonresonant parabolic connections, including the reducible exceptional components.

The compact applications concern elliptic surfaces and minimal δ -regular abelian fibrations. Néron group spaces provide Oka open strata, while singular fibres admit finite Oka filtrations. The same methods prove the Oka property for the Gross–Popescu derived pair. The final section treats relative compactified Jacobians for one-node integral families; their line-bundle loci and closed non-invertible loci are combined by the Oka union theorem.

1 Oka-theoretic mechanisms

1.1 Background and descent

The companion manuscript treats complex manifolds as connected unless explicitly stated otherwise. Its results are applied componentwise below when a manifold is disconnected. It proves that a connected complex manifold is Oka if and only if it is strongly dominable [17, Theorem 1.3]. It also proves that if X is connected, $A \subsetneq X$ is a closed complex analytic subset, and $X \setminus A$ is Oka, then X is Oka [17, Theorem 1.5]. As a consequence, a complex manifold with a finite Oka stratification is Oka [17, Corollary 4.5].

For the finite-stratification consequence, consider a filtration

$$X = X_0 \supset X_1 \supset \cdots \supset X_k = \emptyset$$

whose differences $X_i \setminus X_{i+1}$ are smooth and Oka. With $U_i = X \setminus X_i$, the stratum $X_i \setminus X_{i+1}$ is a closed smooth submanifold of U_{i+1} , and its complement there is U_i . On each connected component of U_{i+1} , either this stratum is the whole component, which is Oka, or it is proper and the Oka

union theorem [17, Theorem 1.5] applies. Starting with the Oka manifold $U_1 = X_0 \setminus X_1$, induction reaches $U_k = X$.

We also use the standard permanence theorems for Oka manifolds. Complex homogeneous manifolds are Oka [20, §5.6]. Holomorphic vector bundles and projective-space bundles over Oka bases are Oka, and the Oka property passes in both directions through unramified coverings and holomorphically locally trivial bundles with Oka fibre [21, Corollary 2.52]. Holomorphic retracts of Oka manifolds are Oka, and complements of algebraic subsets of codimension at least two in affine space are Oka [20, §§5.5–5.6 and 6.4]. Kusakabe's localization theorem applies to covers by Zariski-open Oka subsets [39, Theorem 1.4].

Proposition 1.1. *Suppose $X_1 \subset X_2 \subset \cdots$ are Oka open submanifolds of a complex manifold X and $X = \bigcup_j X_j$. Then X is Oka.*

Proof. To prove the convex approximation property, let $K \subset \mathbb{C}^n$ be compact and convex, and let f be holomorphic on a neighbourhood V of K with values in X . The sets $f^{-1}(X_j)$ form an increasing open cover of K . Compactness gives a finite subcover, and increasingness replaces it by one set $f^{-1}(X_N)$. Replace V by the open neighbourhood $V \cap f^{-1}(X_N)$ of K . Now f takes values in X_N . Convex approximation in the Oka manifold X_N gives entire maps $\mathbb{C}^n \rightarrow X_N$ approximating f uniformly on K . Composing with $X_N \hookrightarrow X$ proves the required approximation in X . The equivalence between convex approximation and the Oka property originates in [19]; we use the formulation [20, Proposition 5.15.1]. In each application below, the sets X_j are open and Oka. $\square$

The parametric Oka theorem, in its form equivalent to the Oka property, is used in the form of [20, Theorem 5.4.4 and Proposition 5.15.1]. Let S be a Stein manifold, let Y be Oka, let P be a compact polyhedron, and let $P_0 \subset P$ be a closed subpolyhedron. A continuous family $f : S \times P \rightarrow Y$ that is holomorphic in S for parameters in P_0 deforms, relative to those parameters, to a family holomorphic in S for every parameter. Approximation on an $\mathcal{O}(S)$ -convex compact set and interpolation on a closed complex subvariety can be retained when the initial data are holomorphic there.

Corollary 1.2. *For every Stein manifold S and Oka manifold Y , the inclusion*

$$\mathrm{Hol}(S, Y) \longrightarrow \mathrm{Cont}(S, Y)$$

is a weak homotopy equivalence in the compact-open topology. If two holomorphic maps $f_0, f_1 : S \rightarrow Y$ are continuously homotopic, then there is a jointly holomorphic map

$$H : S \times \mathbb{C} \rightarrow Y, \quad H(x, 0) = f_0(x), \quad H(x, 1) = f_1(x).$$

If the continuous homotopy fixes a prescribed point of S , then H may be chosen to fix that point for every parameter.

Proof. Surjectivity on path components follows by taking P to be a point. If two holomorphic maps are continuously homotopic, take $P = [0, 1]$ and $P_0 = \{0, 1\}$ to obtain a path through holomorphic maps, proving injectivity on path components. For a based map $S^k \rightarrow \mathrm{Cont}(S, Y)$ whose basepoint is holomorphic, apply the parametric theorem to the adjoint family with $P = S^k$ and P_0 its basepoint. This proves surjectivity on π_k . If a map $S^k \rightarrow \mathrm{Hol}(S, Y)$ is null-homotopic in the continuous mapping space, apply the theorem to its extension over D^{k+1} , with $P_0 = S^k$, to prove injectivity on π_k . The same argument based at any holomorphic map treats every component.

For the second assertion, let f_s be a continuous homotopy from f_0 to f_1 , and put

$$\eta(t) = \frac{|t|}{|t| + |t - 1|}.$$

Then $f_{\eta(t)}(x)$ is a continuous map on $S \times \mathbb{C}$ whose restrictions to $S \times \{0, 1\}$ are holomorphic. Apply parametric interpolation on the Stein manifold $S \times \mathbb{C}$, adding the product of the fixed point with $\mathbb{C}$ to the interpolation subvariety when required. $\square$

The basic principles above also yield the following descent criterion.

Proposition 1.3. *Let $p : Y \rightarrow X$ be a holomorphic map between complex manifolds. Suppose Y is Oka and that, for every $x \in X$, there is $y \in Y$ such that*

$$p(y) = x, \quad dp_y : T_y Y \longrightarrow T_x X \quad \text{is surjective.}$$

Then X is Oka. In particular, every surjective holomorphic submersion from an Oka manifold has an Oka target.

Proof.

Fix $x \in X$, and choose y as in the hypothesis. Work in the connected components containing these points, and write $m = \dim_y Y$, $n = \dim_x X$. Surjectivity implies $m \geq n$.

Since Y is Oka, it is strongly dominable. Thus there is an entire map

$$F : \mathbb{C}^m \longrightarrow Y, \quad F(0) = y, \quad dF_0 : \mathbb{C}^m \xrightarrow{\sim} T_y Y.$$

The linear map $dp_y \circ dF_0$ has rank n . Choose an n -dimensional complex linear subspace $L \subset \mathbb{C}^m$ complementary to its kernel, and choose a linear isomorphism $\iota : \mathbb{C}^n \rightarrow L$. The entire map

$$H = p \circ F \circ \iota : \mathbb{C}^n \longrightarrow X$$

satisfies $H(0) = x$ and

$$dH_0 = dp_y \circ dF_0 \circ \iota,$$

which is an isomorphism. Since x was arbitrary, X is strongly dominable. The strong dominability criterion [17, Theorem 1.3] proves that X is Oka. $\square$

1.2 Complements of product analytic subsets

Theorem 1.4. *Let X and Y be connected positive-dimensional Oka manifolds. Let $A \subsetneq X$ and $B \subsetneq Y$ be closed complex analytic subsets. Then*

$$(X \times Y) \setminus (A \times B)$$

is Oka.

Proof.

Set $n = \dim X$ and $m = \dim Y$. Fix

$$(p, q) \in (X \times Y) \setminus (A \times B).$$

At least one of $p \notin A$ and $q \notin B$ holds. Interchanging the factors if necessary, assume $p \notin A$. Choose entire maps

$$f : \mathbb{C}^n \rightarrow X, \quad g : \mathbb{C}^m \rightarrow Y$$

with $f(0) = p$, $g(0) = q$, and invertible differentials at 0. Such maps exist because X and Y are Oka.

The image of g contains an open neighbourhood of q . A proper analytic subset of a connected complex manifold has empty interior. There is therefore $c \in \mathbb{C}^m$ with $g(c) \notin B$.

The inverse image $Z = f^{-1}(A)$ is a closed analytic subset of the Stein manifold $\mathbb{C}^n$, and $0 \notin Z$. Cartan's theorem A [20, §2.6], applied to its coherent ideal sheaf, gives a holomorphic function $h \in \mathcal{O}(\mathbb{C}^n)$ satisfying

$$h|_Z = 0, \quad h(0) = 1.$$

At 0 the ideal stalk is the whole local ring, so global sections generating this stalk contain a section nonzero at 0; division by its value gives the required normalization.

Define

$$\Phi : \mathbb{C}^n \times \mathbb{C}^m \longrightarrow X \times Y, \quad \Phi(u, v) = (f(u), g(c + h(u)(v - c))).$$

This is entire. If $f(u) \in A$, then $h(u) = 0$, so its second coordinate is $g(c) \notin B$. Thus its image is contained in the asserted complement. At the origin,

$$\Phi(0, 0) = (f(0), g(c - c)) = (p, q).$$

Its differential, with respect to the direct-sum decompositions of the source and target tangent spaces, is

$$d\Phi_{(0,0)} = \begin{pmatrix} df_0 & 0 \\ -dg_0 \circ (c dh_0) & dg_0 \end{pmatrix}.$$

Both diagonal blocks are invertible. Hence this differential is invertible. We have proved strong dominability at the arbitrary point (p, q) , and the strong dominability criterion [17, Theorem 1.3] gives the conclusion. $\square$

Corollary 1.5. *Suppose, in addition, that A and B are smooth Oka submanifolds. Then*

$$\text{Bl}_{A \times B}(X \times Y)$$

is Oka.

Proof.

The blowup of a smooth manifold along a smooth closed centre is smooth. Its exceptional divisor is

$$E = \mathbb{P}(N_{A \times B/(X \times Y)}) = \mathbb{P}(\text{pr}_A^* N_{A/X} \oplus \text{pr}_B^* N_{B/Y}).$$

It is a holomorphically locally trivial projective-space bundle over $A \times B$. The base and fibre are Oka, so E is Oka. The blowup restricts to a biholomorphism outside E , and its complementary open set is the complement in Section 1.2. Apply the Oka union theorem [17, Theorem 1.5] to E . $\square$

Corollary 1.6. *If $D \subset X \times Y$ is finite, then both $(X \times Y) \setminus D$ and the blowup at the distinct points of D are Oka.*

Proof.

Let $A = \text{pr}_X(D)$ and $B = \text{pr}_Y(D)$. They are finite proper analytic subsets, since X, Y have positive dimension. Section 1.2 shows that

$$U = (X \times Y) \setminus (A \times B)$$

is Oka. In the smooth manifold $(X \times Y) \setminus D$, the complement of U is the finite set $(A \times B) \setminus D$. It is a closed zero-dimensional Oka submanifold. Applying the Oka union theorem [17, Theorem 1.5] gives the punctured product. A second application of the same theorem gives the point blowup, whose exceptional divisor is a finite disjoint union of copies of $\mathbb{P}^{n+m-1}$. $\square$

Theorem 1.7. *Let $X_1, \dots, X_s$ be connected positive-dimensional Oka manifolds, and let $A_i \subsetneq X_i$ be closed analytic subsets. Let $\mathcal{H}$ be a finite collection of subsets of $\{1, \dots, s\}$, each of cardinality at least two. Then*

$$\left(\prod_i X_i \right) \setminus \bigcup_{\tau \in \mathcal{H}} \{x : x_i \in A_i \text{ for every } i \in \tau\}$$

is Oka.

Proof. Fix a point $p = (p_i)$ of this complement. Call i good if $p_i \notin A_i$. Every $\tau \in \mathcal{H}$ contains a good index. Reorder the factors so that all good indices precede all other indices. If $i = \max \tau$, there is a good $j_\tau \in \tau \setminus \{i\}$: when i is not good, choose any good index in τ ; when i is good, all indices of τ are good and there is another because $|\tau| \geq 2$.

Choose entire dominating maps $f_i : \mathbb{C}^{n_i} \rightarrow X_i$ with $f_i(0) = p_i$, and choose c_i with $f_i(c_i) \notin A_i$. For every good j , Cartan A [20, §2.6] gives an entire h_j vanishing on $f_j^{-1}(A_j)$ and satisfying $h_j(0) = 1$. Define successively

$$u_i = c_i + (z_i - c_i) \prod_{\tau \in \mathcal{H}: \max \tau = i} h_{j_\tau}(u_{j_\tau}), \quad F(z) = (f_1(u_1), \dots, f_s(u_s)).$$

An empty product is one. Every expression uses only preceding u_j , so all functions are entire. Induction gives $u_i(0) = 0$. The differential of $z \mapsto u$ at zero is block triangular with identity diagonal blocks, because every factor $h_j(u_{j_\tau}(0))$ equals one. Composing with the invertible differentials $df_i(0)$ makes dF_0 invertible.

If $F(z)$ satisfied all the bad conditions for some τ , then $f_{j_\tau}(u_{j_\tau}) \in A_{j_\tau}$, so the corresponding factor vanishes. For $i = \max \tau$, this forces $u_i = c_i$, giving $f_i(u_i) \notin A_i$, a contradiction. Thus F maps to the complement and dominates it at p . Apply the strong dominability criterion [17, Theorem 1.3]. $\square$

1.3 Families and deformation spaces

Lemma 1.8. *Let $p : G \rightarrow B$ be a holomorphic submersion with holomorphic fibrewise group operations and an identity section. The tangent spaces to the fibres along the identity section form a holomorphic vector bundle $\mathfrak{g} \rightarrow B$, and there is a holomorphic map*

$$\text{Exp} : \mathfrak{g} \longrightarrow G$$

whose restriction to each fibre is the exponential map. Its vertical differential along the zero section is the identity.

Proof. A local holomorphic section of $\mathfrak{g}$ determines a vertical left-invariant holomorphic vector field on G . Such a field is complete on every fibre. The holomorphic parameter-dependence theorem for ordinary differential equations makes its complex-time flow holomorphic in the base point, the Lie-algebra vector, and time. Evaluation at time one gives Exp , and the standard initial-value calculation gives the stated differential along the zero section. $\square$

Theorem 1.9. *Let $p : G \rightarrow B$ be a holomorphic submersion equipped with holomorphic fibrewise multiplication, inversion, and an identity section. Suppose every fibre is a connected complex Lie group. Then*

$$G \text{ is strongly dominable} \iff B \text{ is strongly dominable.}$$

Consequently, the strong dominability criterion [17, Theorem 1.3] gives

$$G \text{ is Oka} \iff B \text{ is Oka.}$$

Proof.

Write $b = \dim B$ and $r = \dim G - \dim B$, working componentwise. Let $\mathfrak{g} \rightarrow B$ and $\text{Exp} : \mathfrak{g} \rightarrow G$ be the bundle and map in Lemma 1.8.

Assume B is strongly dominable, and fix $g_0 \in G_{b_0}$. Choose

$$f : \mathbb{C}^b \rightarrow B, \quad f(0) = b_0, \quad df_0 \text{ invertible.}$$

Pull back G and $\mathfrak{g}$ along f . The vector bundle $f^*\mathfrak{g}$ is holomorphically trivial: $\mathbb{C}^b$ is contractible and Stein, so topological triviality becomes holomorphic triviality by the Oka–Grauert principle [27][20, Theorem 5.3.1]. Choose a global holomorphic frame $e_1, \dots, e_r$.

The exponential image in a connected Lie group contains a neighbourhood of its identity. The subgroup generated by this neighbourhood is open. Its cosets are open, so connectedness implies that this subgroup is the whole group. Since the inverse of $\exp(v)$ is $\exp(-v)$, there are $v_1, \dots, v_s \in \mathfrak{g}_{b_0}$ with

$$g_0 = \exp(v_1) \cdots \exp(v_s).$$

Express the v_i in the frame at 0, and extend them by the same constant coefficients to holomorphic sections $v_i(u)$ of $f^*\mathfrak{g}$. Then

$$\sigma(u) = \text{Exp}(v_1(u)) \cdots \text{Exp}(v_s(u))$$

is a holomorphic section of the pulled-back group family with $\sigma(0) = g_0$. Define

$$F : \mathbb{C}^b \times \mathbb{C}^r \rightarrow G, \quad F(u, z) = \sigma(u) \text{Exp} \left(\sum_{i=1}^r z_i e_i(u) \right),$$

where the final map includes the natural projection from the pullback to G . We have $F(0, 0) = g_0$ and $p \circ F(u, z) = f(u)$. Choose a complex-linear splitting of

$$0 \longrightarrow T_{g_0} G_{b_0} \longrightarrow T_{g_0} G \xrightarrow{dp_{g_0}} T_{b_0} B \longrightarrow 0.$$

Relative to the resulting horizontal and vertical coordinates, the derivative has block form

$$dF_{(0,0)} = \begin{pmatrix} df_0 & 0 \\ * & d(L_{g_0})_{e(b_0)} \circ e(0) \end{pmatrix}, \quad e(0)(z) = \sum_{i=1}^r z_i e_i(0).$$

The upper-left block is invertible by the choice of f . The lower-right block is the composite of the frame isomorphism with the derivative of a biholomorphic left translation and is therefore invertible. In particular, if $dF_{(0,0)}(\xi, \eta) = 0$, the first row gives $\xi = 0$, and the second gives $\eta = 0$. Source and target both have dimension $b + r$, so the derivative is an isomorphism. This proves strong dominability of G .

Conversely, compose a dominating entire map at the identity over b_0 with p , and restrict to a linear complement of the differential kernel as in Section 1.1. This gives strong dominability at b_0 . Alternatively, for the Oka statement in this direction, the identity section and p make B a holomorphic retract of G . Applying the strong dominability criterion [17, Theorem 1.3] to the forward implication completes the proof. $\square$

Corollary 1.10. *Let $p : G \rightarrow B$ be a smooth holomorphic family of complex Lie groups whose identity-component family $G^0 \rightarrow B$ has connected fibres. Suppose B is Oka and $G \setminus G^0$ is a finite disjoint union of closed connected components of special fibres. Then G is Oka.*

Proof. Theorem 1.9 makes G^0 Oka. Every additional component is a torsor under a connected complex Lie group, hence is a homogeneous Oka manifold. The Oka union theorem [17, Theorem 1.5] adds these closed components to G^0 . In an algebraic group scheme over a curve with connected generic fibre, the additional components are precisely such a finite union. $\square$

The same mechanisms apply to the deformation to the normal cone in the standard form of [22, §5.1].

Let B be a complex manifold and $J \subset B$ a smooth closed submanifold. The deformation space of B along J is

$$\mathcal{D}_J(B) = \mathrm{Bl}_{J \times \{0\}}(B \times \mathbb{C}) \setminus \widetilde{B \times \{0\}}.$$

Theorem 1.11. *If B and J are Oka, then $\mathcal{D}_J(B)$ is Oka.*

Proof. The central fibre is the normal bundle $N_{J/B}$, while its complement is $B \times \mathbb{C}^*$. Both are Oka. The total space is smooth: in coordinates with $J = \{x_1 = \cdots = x_c = 0\}$, the blowup charts meeting the central fibre have coordinates

$$(u_1, \dots, u_c, x_{c+1}, \dots, x_N, t), \quad x_i = tu_i.$$

Thus the central fibre is a closed smooth Oka submanifold with Oka complement, and the Oka union theorem [17, Theorem 1.5] applies. $\square$

When $B = \mathbb{A}^N$ and $I = I(J)$, the coordinate ring is the extended Rees algebra

$$\mathbb{C}[x_1, \dots, x_N, t, I/t] \subset \mathbb{C}[x_1, \dots, x_N, t, t^{-1}].$$

Hence the construction produces smooth affine Oka manifolds from smooth affine Oka centres.

2 Stable and parabolic bundles on curves

The theory of stable bundles on curves and their moduli spaces begins with the Narasimhan–Seshadri correspondence [42] and Seshadri’s construction of the moduli space [51].

Let C be a smooth projective connected curve of genus $g \geq 2$, let $r \geq 2$, and let $\xi \in \mathrm{Pic}^d(C)$. Write

$$M_C(r, \xi) = \mathrm{SU}_C^s(r, \xi)$$

for the coarse moduli space of stable rank- r vector bundles with determinant isomorphic to ξ .

Theorem 2.1. *If*

$$\gcd(r, d) = 1 \quad \text{or} \quad (r, g) \neq (2, 2),$$

then $M_C(r, \xi)$ is an Oka manifold. The moduli space $U_C^s(r, d)$ with variable determinant is Oka in the same range. When $\gcd(r, d) = 1$, stability equals semistability, so $\mathrm{SU}_C(r, \xi)$ is a smooth projective Oka manifold.

Choose a line bundle L of sufficiently negative degree l , and put

$$M = \xi \otimes L^{-r}, \quad e = d - rl.$$

We require

$$e - rg \geq 2, \quad \frac{e}{r} > 2g - 1. \quad (2.1)$$

Let

$$V = H^1(C, M^{-1})^{\oplus(r-1)}.$$

Since M and $\mathcal{O}_C^{r-1}$ are fixed, V carries a universal extension; see Bertram [8, §2] in rank two and Heinloth [32, Lectures 1–3] in general:

$$0 \longrightarrow \mathcal{O}_C^{r-1} \xrightarrow{i_v} E_v \longrightarrow M \longrightarrow 0. \quad (2.2)$$

For example, choose a finite affine cover of C , choose Čech cocycles representing a basis of $H^1(M^{-1})$, and use their linear combinations as the upper-right blocks of the transition matrices of ((2.2)). This constructs an algebraic family on $C \times V$. Its determinant has the natural specified isomorphism $\det E_v \simeq M$.

Let $V^s \subset V$ be the Zariski-open locus on which E_v is stable. We will prove two separate assertions:

$$\text{codim}_V(V \setminus V^s) \geq 2, \quad V^s \longrightarrow M_C(r, \xi), \quad v \longmapsto [E_v \otimes L], \text{ is a surjective submersion.} \quad (2.3)$$

The first assertion makes the source Oka, and the strong dominability criterion [17, Theorem 1.3] applied to the second gives the target.

Define

$$B_{\text{ng}} = \{v \in V : \text{there is } p \in C \text{ with } H^1(C, E_v(-p)) \neq 0\}.$$

For every p , the degree assumption gives $H^1(M(-p)) = 0$. The cohomology sequence of ((2.2)), tensored by $\mathcal{O}_C(-p)$, therefore identifies the condition in the definition of B_{ng} with failure of surjectivity of

$$\delta_{v,p} : H^0(M(-p)) \longrightarrow H^1(\mathcal{O}_C(-p))^{r-1}.$$

By Serre duality, a failure of surjectivity is witnessed by a nonzero tuple

$$\alpha = (\alpha_1, \dots, \alpha_{r-1}) \in H^0(K_C(p))^{r-1} = H^0(K_C)^{r-1}.$$

The equality follows from the inclusion $H^0(K_C) \subset H^0(K_C(p))$ and Riemann–Roch, which gives dimension g for both spaces.

For fixed p and $[\alpha]$, the equation $\alpha \circ \delta_{v,p} = 0$ is linear in v . Its transpose under Serre duality is

$$\mu_\alpha : H^0(M(-p)) \longrightarrow H^0(MK_C)^{r-1}, \quad s \longmapsto (s\alpha_1, \dots, s\alpha_{r-1}).$$

At least one α_i is nonzero. Since C is integral, multiplication by that differential is injective on $H^0(M(-p))$. Thus μ_α is injective. Riemann–Roch gives

$$h^0(M(-p)) = e - g,$$

so the linear conditions on V have codimension exactly $e - g$.

The witnesses form the proper variety

$$C \times \mathbb{P}(H^0(K_C)^{r-1}),$$

of dimension $(r-1)g$. The incidence conditions are algebraic; projection to V is proper because the witness space is proper. Consequently B_{ng} is closed and

$$\text{codim}_V B_{\text{ng}} \geq (e - g) - (r-1)g = e - rg \geq 2. \quad (2.4)$$

Set $V^{\text{reg}} = V \setminus B_{\text{ng}}$. The sequence

$$0 \rightarrow E_v(-p) \rightarrow E_v \rightarrow (E_v)_p \rightarrow 0$$

shows that every E_v , $v \in V^{\text{reg}}$, is globally generated and has $H^1(E_v) = 0$.

If E_v is stable, then $H^1(E_v(-p)) = 0$ for every p : a nonzero dual map $E_v \rightarrow K_C(p)$ would give a rank-one quotient image of degree at most $2g - 1$, whereas stability requires every proper quotient to have slope larger than $e/r > 2g - 1$. Hence

$$V^s \subset V^{\text{reg}}. \quad (2.5)$$

Use the determinant-rigidified stack

$$\mathcal{B}_{r,M} = \{(E, \det E \xrightarrow{\sim} M)\}.$$

Its dimension is $(r^2 - 1)(g - 1)$ [32, Lectures 1–3]. In the stack-dimension computations below, the Picard **stack** has dimension $g - 1$, while the Picard variety has dimension g .

The family ((2.2)) gives a smooth morphism

$$V^{\text{reg}} \longrightarrow \mathcal{B}_{r,M}. \quad (2.6)$$

Here is the infinitesimal lifting argument. Consider a square-zero deformation of E_v with its determinant isomorphism, and the $r - 1$ sections defining i_v . Each section-lifting obstruction lies in $H^1(E_v)$, tensored with the square-zero ideal; these groups vanish on V^{reg} . The sections therefore lift simultaneously. Being a subbundle with locally free quotient is an open condition on the resulting matrix, so the lifted sections still give a rank- $(r - 1)$ subbundle. The determinant isomorphism identifies the quotient line with M . We have thus lifted the entire extension with its endpoint identifications, which is a point of the affine extension family. This proves formal smoothness. Finite presentation then gives smoothness.

The corresponding tangent calculation is also explicit. Restriction to the first $r - 1$ columns gives an exact sequence of vector bundles

$$0 \rightarrow \text{Hom}(M, \mathcal{O}_C^{r-1}) \rightarrow \text{End}_0(E_v) \rightarrow \text{Hom}(\mathcal{O}_C^{r-1}, E_v) \rightarrow 0. \quad (2.7)$$

Locally split ((2.2)). Any first $r - 1$ columns of an $r \times r$ matrix can be prescribed; the last diagonal entry is uniquely determined by trace zero. The kernel consists of the first $r - 1$ entries of the last column, which is exactly $\text{Hom}(M, \mathcal{O}_C^{r-1})$. Since $H^1(E_v) = 0$, the cohomology sequence makes the Kodaira–Spencer map onto $H^1(\text{End}_0 E_v)$ surjective.

A nonstable bundle E has a saturated subbundle $F \subset E$ of rank s , degree f , where

$$1 \leq s \leq r - 1, \quad q = r - s, \quad A = rf - se \geq 0.$$

Put $Q = E/F$. Consider the stack of sequences

$$0 \rightarrow F \rightarrow E \rightarrow Q \rightarrow 0, \quad \det F \otimes \det Q \simeq M.$$

The stack of pairs (F, Q) with this determinant identification has dimension

$$(s^2 + q^2 - 1)(g - 1).$$

Indeed, the product of the two unrestricted bundle stacks has dimension $(s^2 + q^2)(g - 1)$, and fixing their product determinant subtracts the dimension $g - 1$ of the Picard stack.

For each pair, the stack of extensions has relative dimension

$$\dim \text{Ext}^1(Q, F) - \dim \text{Hom}(Q, F) = -\chi(Q^* \otimes F).$$

Now $Q^* \otimes F$ has rank sq and degree

$$qf - s(e - f) = rf - se = A.$$

Therefore its extension-stack contribution is $sq(g - 1) - A$, and the total extension stack has dimension

$$(r^2 - sq - 1)(g - 1) - A.$$

Forgetting the subbundle is representable: after fixing E , its fibre is the relevant locally free quotient locus in a Quot scheme. The dimension of its image cannot exceed that of the source. Comparison with $\dim \mathcal{B}_{r,M}$ gives the lower bound

$$\text{codim}(\text{nonstable type } s, f) \geq sq(g - 1) + A. \quad (2.8)$$

This includes the slope-equality case.

Only finitely many degrees f occur in the extension family. A destabilizing F has $f \geq \lceil se/r \rceil > 0$, so it cannot lie in $\mathcal{O}_C^{r-1}$: the determinant of a subsheaf of a trivial bundle has degree at most zero. Its image in M is $M(-D)$ for an effective divisor D . The kernel is a subsheaf of $\mathcal{O}_C^{r-1}$, again of nonpositive degree, and hence

$$\lceil se/r \rceil \leq f \leq e. \quad (2.9)$$

There are finitely many s, f , and the corresponding relative Quot schemes over the finite-type space V are of finite type; their images therefore form a finite union.

Smooth pullback along ((2.6)) preserves the codimension bound. If $\gcd(r, e) = \gcd(r, d) = 1$, then A cannot be zero for $0 < s < r$, so

$$sq(g - 1) + A \geq (r - 1)(g - 1) + 1 \geq 2.$$

For $\gcd(r, d) > 1$, the same expression is at least $(r - 1)(g - 1)$; it is at least two when $(r, g) \neq (2, 2)$. Together with ((2.4)), stability openness gives

$$\text{codim}_V(V \setminus V^s) \geq 2$$

in precisely the range stated in Theorem 2.1.

Let E be any stable bundle with determinant ξ . The bundle $E' = E \otimes L^{-1}$ has slope $e/r > 2g - 1$, so the quotient-slope argument already used proves that E' is globally generated and $H^1(E') = 0$.

We justify the choice of $r - 1$ everywhere independent global sections. At any $p \in C$, evaluation

$$H^0(E')^{r-1} \longrightarrow \text{Mat}_{r \times (r-1)}(\mathbb{C})$$

is surjective. Matrices of rank at most $r - 2$ have codimension two. The incidence obtained by allowing p to vary therefore has codimension at least one in the section-tuple space. Its projection is closed because C is projective. A tuple outside that image has no rank drop anywhere. It defines

$$0 \rightarrow \mathcal{O}_C^{r-1} \rightarrow E' \rightarrow \det E' = M \rightarrow 0,$$

so E' occurs in ((2.2)).

Consequently $V^s \rightarrow M_C(r, \xi)$ is surjective. It is a holomorphic submersion by ((2.6)): on the stable locus the determinant-rigidified stack is a μ_r -gerbe over its smooth coarse moduli space. Over $\mathbb{C}$, μ_r is finite étale, and this gerbe has no infinitesimal directions; thus the coarse classifying map retains the surjective differential.

The affine codimension-two complement theorem makes V^s Oka. Regular-preimage descent and the strong dominability criterion [17, Theorem 1.3] prove Theorem 2.1 for fixed determinant.

For variable determinant, the morphism

$$SU_C^s(r, \xi) \times \text{Pic}^0(C) \longrightarrow U_C^s(r, d), \quad (E, N) \longmapsto E \otimes N, \quad (2.10)$$

is the base change of multiplication by r on $\mathrm{Pic}^0(C)$ along the determinant map, after identifying $\mathrm{Pic}^d(C)$ with $\mathrm{Pic}^0(C)$ by ξ . It is a finite étale surjection. The source is Oka, because a Jacobian is a complex Lie group. Covering descent proves the remaining assertion.

We now add parabolic data at marked points. Parabolic bundles and their stability condition were introduced by Mehta–Seshadri [40, §§1–2].

Fix distinct marked points $p_1, \dots, p_m$ on a smooth projective connected curve C . At p_i , fix a flag type with positive graded dimensions

$$n_{i1}, \dots, n_{it_i}, \quad \sum_h n_{ih} = r,$$

and rational weights

$$0 \leq \alpha_{i1} < \dots < \alpha_{it_i} < 1.$$

We use descending flags, so the deeper graded pieces have the larger weights. A flag type is nontrivial when $t_i \geq 2$. The parabolic degree is

$$\mathrm{pardeg}(E_*) = \deg E + \sum_{i,h} n_{ih} \alpha_{ih}.$$

Stability means that every nonzero proper saturated subbundle, equipped with the induced flags and weights, has strictly smaller parabolic slope.

Theorem 2.2. *Suppose $g(C) \geq 2$, $r \geq 2$, and at least one marked point has a nontrivial flag type. For every integer d , every $\xi \in \mathrm{Pic}^d(C)$, and every weight system as above, the smooth coarse moduli space*

$$\mathcal{P}^s(C, \mathbf{p}; r, \xi, \mathbf{n}, \boldsymbol{\alpha})$$

of stable parabolic bundles is Oka. The variable-determinant stable moduli space is Oka as well.

Whenever semistability equals stability, the full projective parabolic moduli space is a smooth projective Oka manifold. This applies in particular to full flags with generic rational weights.

Choose L, M, e, V as in Section 2, increasing e so that

$$e - rg \geq 2, \quad \frac{e}{r} > 2g - 1 + m. \quad (2.11)$$

Let $\mathcal{E}$ be the universal extension on $C \times V$. Form the fibre product of flag bundles

$$W = \prod_{i=1}^m {}_V \mathrm{Flag}_{\mathbf{n}_i}(\mathcal{E}|_{\{p_i\} \times V}).$$

For each marked point, restriction of the universal extension gives

$$0 \rightarrow \mathcal{O}_V^{r-1} \rightarrow \mathcal{E}_{p_i} \rightarrow M_{p_i} \otimes \mathcal{O}_V \rightarrow 0.$$

Since V is affine, $H^1(V, \mathcal{O}_V) = 0$, and this extension of bundles on V splits algebraically. Hence

$$W \simeq V \times \prod_i \mathrm{Flag}_{\mathbf{n}_i}(\mathbb{C}^r). \quad (2.12)$$

Put $W^{\mathrm{reg}} = W \times_V V^{\mathrm{reg}}$. The complement has codimension at least two by ((2.4)). The map from W^{reg} to the determinant-rigidified quasi-parabolic bundle stack is smooth. To verify this, start with a deformation of a flagged bundle. Lift the $r - 1$ sections using $H^1(E) = 0$, exactly as in Section 2. Transport the given flags along the resulting identification of the lifted bundle with the extension family. Because W parametrizes all flags of the prescribed types, these flags give the required lift in W . Thus the section-lifting proof applies with the flag data included.

Every parabolic stable object lies in this regular atlas. For a proper quotient Q_* of a parabolic stable bundle E_* , stability gives

$$\text{par } \mu(Q_*) > \text{par } \mu(E_*).$$

The sum of the average quotient weights over the m marked points is less than m , while the corresponding sum for E_* is nonnegative. Therefore

$$\mu(Q) > \frac{e}{r} - m. \quad (2.13)$$

The same weaker lower bound is satisfied by $\mu(E)$ itself. Consequently

$$\mu_{\min}(E) > \frac{e}{r} - m > 2g - 1.$$

There can be no nonzero map $E \rightarrow K_C(p)$ for any p , because its image would be a rank-one quotient of degree at most $2g - 1$. Thus $H^1(E(-p)) = 0$ for all p , and E is globally generated with $H^1(E) = 0$. The rank-drop incidence in Section 2 supplies $r - 1$ everywhere independent sections. After choosing them, and choosing the given flags in their fibres, the object is represented by a point of W^{reg} .

Fix a saturated subbundle $F \subset E$ of rank s , degree f , and set

$$q = r - s, \quad A = rf - se \in \mathbb{Z}.$$

At one marked point, omit the index i , and let

$$a_h = \dim \text{gr}_h(F), \quad b_h = n_h - a_h.$$

Then $\sum_h a_h = s$ and $\sum_h b_h = q$.

The locus of flags having this relative position with a fixed s -plane has codimension

$$c(\mathbf{a}) = \sum_{u>v} a_u b_v. \quad (2.14)$$

Fix the flag instead and vary the s -plane. Choose its graded subspaces in each block; these choices have dimension $\sum_h a_h b_h$. In echelon form, the additional graph entries run from a selected vector in block u to an unselected vector in a deeper block $v > u$, contributing $\sum_{u<v} a_u b_v$ parameters. Thus the relative-position stratum in $\text{Gr}(s, r)$ has dimension

$$\sum_h a_h b_h + \sum_{u<v} a_u b_v.$$

Subtracting this from $sq = \sum_{u,v} a_u b_v$ gives ((2.14)). The same codimension holds when the subspace is fixed and the flag varies: the incidence of subspace-flag pairs is homogeneous over the Grassmannian, and subtracting its base dimension gives exactly this fibre codimension.

The weight contribution to the stability inequality is

$$\begin{aligned} w(\mathbf{a}) &= r \sum_h a_h \alpha_h - s \sum_h n_h \alpha_h \\ &= \sum_{u,v} a_u b_v (\alpha_u - \alpha_v). \end{aligned} \quad (2.15)$$

Subtracting ((2.15)) from ((2.14)) yields the exact identity

$$\begin{aligned} c(\mathbf{a}) - w(\mathbf{a}) &= \sum_{u>v} a_u b_v (1 - \alpha_u + \alpha_v) \\ &\quad + \sum_{u<v} a_u b_v (\alpha_v - \alpha_u). \end{aligned} \quad (2.16)$$

Every displayed coefficient is strictly positive because the distinct weights increase and have width less than one.

For a nontrivial flag, ((2.16)) is strictly positive. Indeed, if it were zero, every pair with $a_u > 0$ and $b_v > 0$ would have $u = v$. Both collections are nonempty because $s, q > 0$. It follows that all a_u and all b_v are supported in a single common block. Every other graded multiplicity would then be $n_h = a_h + b_h = 0$, contradicting the positive multiplicities in a flag with at least two blocks. A trivial flag contributes zero.

For a full flag, write the selected weight indices as

$$I = \{i_1 < \cdots < i_s\} \subset \{1, \dots, r\}.$$

The same formulas become

$$c(I) = \sum_{k=1}^s (i_k - k), \quad w(I) = r \sum_{j \in I} \alpha_j - s \sum_{j=1}^r \alpha_j. \quad (2.17)$$

For rank two, the generic line receives the lower weight. Equality with the distinguished flag line gives the higher weight and is a codimension-one condition, which also fixes the direction of the indexing convention.

The subbundle F destroys parabolic stability exactly when

$$A + \sum_i w_i \geq 0. \quad (2.18)$$

At least one nontrivial flag makes $\sum_i (c_i - w_i) > 0$. Combining this with ((2.18)), and using that $A + \sum_i c_i$ is an integer, gives

$$A + \sum_i c_i \geq 1. \quad (2.19)$$

The extension-stack computation in Section 2 applies to every degree f . Adding the relative flag conditions gives the following codimension bound in the quasi-parabolic bundle stack:

$$\begin{aligned} \text{codim}(s, f, \mathbf{a}_i) &\geq sq(g-1) + A + \sum_i c_i \\ &\geq sq(g-1) + 1 \\ &\geq (r-1)(g-1) + 1 \geq 2. \end{aligned} \quad (2.20)$$

Thus the Schubert cost compensates for every possible weight contribution, independently of the chamber.

The type set is finite on W . There are finitely many ranks and relative-position arrays. Since $|w_i| < sq$, ((2.18)) implies

$$f > \frac{se - msq}{r} \geq s \left(\frac{e}{r} - m \right) > 0.$$

A subbundle of positive degree cannot lie in $\mathcal{O}_C^{r-1}$, so the projection to M gives $f \leq e$, as in ((2.9)). The relative Quot schemes with these finitely many Hilbert polynomials are of finite type. Smooth pullback preserves ((2.20)), and $W \setminus W^{\text{reg}}$ already has codimension at least two. Parabolic stability is Zariski open for fixed rational weights. Therefore, if $W^{\text{par},s}$ denotes the stable locus,

$$\text{codim}_W(W \setminus W^{\text{par},s}) \geq 2. \quad (2.21)$$

Each partial flag variety has a finite Zariski-open cover by affine big cells: translates of its open big cell cover the homogeneous variety, and quasi-compactness selects finitely many translates. By ((2.12)), W has a finite cover by affine spaces. In each such chart, $W^{\text{par},s}$ is the complement of a closed algebraic subset of codimension at least two, hence Oka. Zariski localization then makes $W^{\text{par},s}$ Oka.

The classifying map

$$W^{\text{par},s} \longrightarrow \mathcal{P}^s(C, \mathbf{p}; r, \xi, \mathbf{n}, \boldsymbol{\alpha})$$

is surjective by Section 2 and is a submersion by the flag-compatible section-lifting argument. A stable parabolic bundle has only scalar parabolic endomorphisms; after specifying the determinant isomorphism, its stabilizer is μ_r . The stable coarse moduli space is smooth because the obstruction group is a second cohomology group of the parabolic endomorphism sheaf on a curve and therefore vanishes. Passing from the stable stack to its coarse moduli space introduces no infinitesimal obstruction. Regular-preimage descent proves Theorem 2.2.

Twisting by $N \in \text{Pic}^0(C)$ preserves all flag types, weights, and stability inequalities. The analogue of ((2.10)) is again the base change of $[r]$ on the Jacobian. This proves the variable-determinant assertion.

Theorem 2.3. *Let C be a smooth projective curve of genus $g = 0$ or 1 , let $r \geq 2$, and fix m distinct marked points, any degree d , and any determinant ξ . At every marked point take a full flag with weights*

$$\alpha_{ij} = \frac{j-1}{r}, \quad j = 1, \dots, r.$$

If

$$(r-1) \left(g - 1 + \frac{m}{2} \right) > 1, \quad (2.22)$$

then the stable parabolic moduli space is nonempty, smooth, and Oka, both with fixed and with variable determinant. Moreover, there are arbitrarily close generic rational weight systems for which the entire nonempty semistable moduli space is stable, smooth, projective, and Oka.

The numerical sufficient ranges are:

Genus	Rank 2	Rank 3	Rank $r \geq 4$
0	$m \geq 5$	$m \geq 4$	$m \geq 3$
1	$m \geq 3$	$m \geq 2$	$m \geq 1$

Proof. Fix a rank s , write $q = r - s$, and use the full-flag indices I . Formula ((2.17)) gives

$$c(I) = \sum_{j \in I} j - \frac{s(s+1)}{2}, \quad w(I) = \sum_{j \in I} j - \frac{s(r+1)}{2}.$$

Hence the cancellation is exact and independent of the relative position:

$$c(I) - w(I) = \frac{sq}{2}. \quad (2.23)$$

A parabolic destabilizer satisfies ((2.18)). Since $A + \sum_i c(I_i)$ is an integer,

$$A + \sum_i c(I_i) \geq \left\lceil \frac{msq}{2} \right\rceil.$$

The same extension-stack computation therefore gives

$$\text{codim}(\text{bad type}) \geq sq(g-1) + \left\lceil \frac{msq}{2} \right\rceil = \left\lceil sq \left(g - 1 + \frac{m}{2} \right) \right\rceil. \quad (2.24)$$

Under ((2.22)), the coefficient $g - 1 + m/2$ is positive and the smallest value of $s(r - s)$ is $r - 1$. Thus every number in ((2.24)) is at least two.

The extension atlas still works in these genera, with one modification in genus zero. If $C = \mathbb{P}^1$ and $e \geq 0$, then for every point p ,

$$H^1(\mathcal{O}_C(-p)) = 0, \quad H^1(M(-p)) = 0.$$

The cohomology sequence of ((2.2)) shows $H^1(E_v(-p)) = 0$ for every v , so $V^{\text{reg}} = V$. In genus zero the dual-witness space vanishes. In genus one, the witness argument in Section 2 applies with $h^0(K_C) = 1$, and choosing $e - r \geq 2$ makes its exceptional set have codimension at least two.

In either genus choose e large enough that

$$\frac{e}{r} > m, \quad \frac{e}{r} > 2g - 1 + m,$$

and also $e - r \geq 2$ in genus one. The quotient-slope bound ((2.13)), the nowhere-rank-drop section incidence, the lifting proof, and the finite interval $0 < f \leq e$ now apply without change to their hypotheses. Equations ((2.21))–((2.24)) show that the complement of the stable locus in the nonempty irreducible parameter space W has codimension at least two. This proves nonemptiness and the Oka property.

To obtain generic weights arbitrarily close to the displayed system, let $\mathcal{U}$ be any neighbourhood of it in the full-flag weight space. For the finitely many ranks and relative positions, the real bounds

$$sq(g - 1) + \sum_i (c_i - w_i) > 1$$

have a uniform positive margin at the displayed weights. Choose $\varepsilon > 0$ so small that

$$\alpha_{ij}^\varepsilon = \frac{j - 1}{r} + \varepsilon$$

lies in the interior of the weight space and belongs to $\mathcal{U}$. A common translation at each marked point does not affect slope comparisons. By continuity, a neighbourhood $\mathcal{V}$ of α^ε can be chosen inside $\mathcal{U}$ so that every one of the preceding bounds remains greater than one throughout $\mathcal{V}$; integrality then again gives codimension at least two. Full-flag stability walls are proper affine hyperplanes: for a proper rank s , the weight coefficients are $r - s$ at selected indices and $-s$ at unselected ones, so they do not vanish identically. Only finitely many degrees can enter a wall because the weights are bounded. Rational points outside this finite union of hyperplanes are dense in $\mathcal{V}$. Choosing one gives a generic rational weight system in $\mathcal{U}$. For these weights semistability equals stability, and the projective parabolic GIT construction [54, §5] gives the claimed compact examples. □

The moduli construction and parabolic stack stratification are developed in Mehta–Seshadri [40, §§1–4], Sun [54, §5], and Fernández Herrero [33, §§7–9]. The codimension estimates used here were derived in Section 2 with the stated flag and weight conventions.

3 Holomorphic mapping and embedding spaces

For curves, the topology of holomorphic mapping spaces was initiated by Segal’s comparison of rational and continuous maps [50]. Aumonier gives the higher-dimensional projective-space framework closest to the mapping-space applications below [3]. Here the Oka property is obtained from explicit entire families of generating sections.

3.1 Maps defined by sections

Let X be a fixed compact complex manifold of dimension $k \geq 1$, let B be an Oka manifold of dimension b , and let

$$\mathcal{F} \longrightarrow X \times B$$

be a holomorphic vector bundle of rank r . Write $p : X \times B \rightarrow B$. Assume that

$$V = p_* \mathcal{F}$$

is a holomorphic vector bundle of rank R , that formation of V commutes with base change, and that the evaluation morphism

$$p^*V \longrightarrow \mathcal{F}$$

is surjective.

Let

$$\text{Gen}_N(\mathcal{F}/B) \subset V^{\oplus N}$$

be the open set of tuples that generate $\mathcal{F}_b$ at every point of X .

Theorem 3.1. *If*

$$N \geq r + k + \max\{1, b\}, \quad (3.1)$$

then $\text{Gen}_N(\mathcal{F}/B)$ is strongly dominable and hence is Oka by the strong dominability criterion [17, Theorem 1.3].

Fix a point

$$p_0 = (b_0, s_1^0, \dots, s_N^0) \in \text{Gen}_N(\mathcal{F}/B).$$

For $x \in X$, the tuple gives a surjection

$$q_x : \mathbb{C}^N \longrightarrow (\mathcal{F}_{b_0})_x.$$

For a nonzero linear functional $\ell \in (\mathbb{C}^N)^*$, restriction to $H = \ker \ell$ fails to be surjective precisely when

$$\ell \in \text{im}(q_x^*). \quad (3.2)$$

Indeed, failure means that a nonzero functional on $(\mathcal{F}_{b_0})_x$ annihilates $q_x(H)$; its pullback is a nonzero multiple of ℓ . The converse is immediate.

Consequently the bad points $[\ell] \in \mathbb{P}((\mathbb{C}^N)^*)$ are the image of

$$\mathbb{P}(\mathcal{F}_{b_0}^*) \longrightarrow \mathbb{P}((\mathbb{C}^N)^*).$$

This is a closed analytic set because X is compact. Its dimension is at most $k + r - 1$. By ((3.1)),

$$k + r - 1 < N - 1,$$

so the bad set is proper.

Choose a basis $\ell_1, \dots, \ell_N$ of $(\mathbb{C}^N)^*$ with every projective point outside this set. Such a basis can be chosen successively: at each stage one avoids a proper analytic set and a proper linear subspace. In the dual basis of $\mathbb{C}^N$, deletion of any one of the N sections leaves a generating tuple.

A fixed change of basis in GL_N is a global holomorphic automorphism of the generation space. It is therefore enough to prove strong dominability after making this change, and then compose with its inverse.

Because B is Oka, choose an entire map

$$\varphi : \mathbb{C}^b \longrightarrow B, \quad \varphi(0) = b_0, \quad d\varphi_0 \text{ an isomorphism.}$$

The implication Oka $\Rightarrow$ strong dominability supplies φ ; at the end, the converse implication in the strong dominability criterion [17, Theorem 1.3] converts the constructed dominating maps into the Oka property.

By the Oka–Grauert principle [27][20, Theorem 5.3.1], φ^*V is holomorphically trivial. Choose an entire frame $e_1(u), \dots, e_R(u)$, and write the initial sections in this frame as vectors $a_j = (a_{ja}) \in \mathbb{C}^R$. For $b \geq 1$, introduce

$$M = NbR$$

complex parameters $c_{j\ell a}$ and set

$$s_j(u; \mathbf{c}) = \sum_{a=1}^R \left(a_{ja} + \sum_{\ell=1}^b c_{j\ell a} u_\ell \right) e_a(u). \quad (3.3)$$

For every choice of parameters this agrees with the prescribed tuple at $u = 0$.

On

$$X \times (\mathbb{C}^b \setminus \{0\}) \times \mathbb{C}^M$$

consider the evaluation matrix with columns $s_j(u; \mathbf{c})(x)$. Its derivative in the $\mathbf{c}$ directions is surjective onto the space of $r \times N$ matrices. Indeed, some $u_\ell \neq 0$, and evaluation of the frame spans $(\mathcal{F}_{\varphi(u)})_x$.

The locus of matrices of rank t is smooth of codimension

$$(r - t)(N - t).$$

Here is the local calculation. On the open set where a specified $t \times t$ minor is invertible, write the matrix in blocks as

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A \in \mathrm{GL}_t, \quad D \in \mathrm{Mat}_{(r-t) \times (N-t)}.$$

Multiplying on the left and right by invertible block matrices transforms it into

$$\begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

Thus rank exactly t is the equation $D = CA^{-1}B$ in this chart. Its $(r - t)(N - t)$ scalar equations have independent differentials in the entries of D . The case $t = 0$ is the single zero matrix. These charts cover every rank stratum, proving both smoothness and the codimension formula.

Among the non-surjective ranks $0 \leq t < r$, the smallest codimension is $N - r + 1$. The inverse image of each rank stratum is therefore a smooth complex manifold of dimension at most

$$M + k + b - (N - r + 1) < M. \quad (3.4)$$

The strict inequality follows from $N \geq r + k + b$.

The variable u is noncompact, so we work with a countable atlas of each second-countable incidence stratum. In each chart its real dimension is less than $2M$, so Sard's theorem implies that the image in $\mathbb{C}^M$ has Lebesgue measure zero. Taking the countable union of charts and the finite union of rank strata still gives a null set.

Choose $\mathbf{c}$ outside this entire image. Then the tuple ((3.3)) generates for every $u \neq 0$, and it generates at $u = 0$ by construction. Thus there is an entire section

$$\sigma : \mathbb{C}^b \longrightarrow \varphi^* \mathrm{Gen}_N(\mathcal{F}/B)$$

through p_0 . Its projection to the base is the identity.

If $b = 0$, take σ to be the prescribed tuple.

Use the chosen frame to identify the pullback generation space with an open subset

$$\tilde{G} \subset \mathbb{C}^b \times (\mathbb{C}^R)^N.$$

For each j , let

$$A_j \subset T_j := \mathbb{C}^b \times (\mathbb{C}^R)^{N-1}$$

be the set where the sections other than the j th fail to generate. The rank-drop incidence in $X \times T_j$ is closed analytic, and its projection is proper because X is compact. Remmert's proper

mapping theorem [47][20, Chapter 2] therefore makes A_j a closed analytic subset of the Stein manifold T_j .

By Section 3.1, the point given by the initial remaining sections is outside A_j . The ideal-sheaf form of Cartan's theorem A [20, §2.6] supplies

$$h_j \in \mathcal{O}(T_j), \quad h_j|_{A_j} = 0, \quad h_j(p_0) \neq 0.$$

The complete vector fields below are elementary shears of the kind underlying the Andersén–Lempert method [2]. For $1 \leq a \leq R$, define

$$V_{ja} = h_j(u, s_j) \frac{\partial}{\partial s_{ja}}.$$

Its complete flow is

$$\Phi_{ja}^t : s_j \mapsto s_j + t h_j(u, s_j) e_a, \quad (3.5)$$

with all other sections and u fixed.

The coefficient is independent of s_j , so this formula is defined for every $t \in \mathbb{C}$, and the inverse is obtained by replacing t with $-t$. It preserves $\tilde{G}$: if the other sections generate, changing s_j arbitrarily is harmless; if they do not generate, then $h_j = 0$ and the tuple does not move.

At p_0 , the NR vector fields V_{ja} span the full vertical tangent space.

Order these fields as $V_1, \dots, V_{NR}$ and define

$$F(u, t) = \Phi_{NR}^{t_{NR}} \circ \dots \circ \Phi_1^{t_1}(\sigma(u)).$$

This is an entire map

$$\mathbb{C}^{b+NR} \longrightarrow \tilde{G}.$$

At $(u, t) = (0, 0)$, the u directions project isomorphically to the base, and the time directions span the entire vertical tangent space. Hence $dF_{(0,0)}$ is an isomorphism.

More explicitly, use the frame coordinates on $\tilde{G}$ and order the time variables by (j, a) . Since every flow at time zero is the identity, the chain rule gives

$$dF_{(0,0)} = \begin{pmatrix} I_b & 0 \\ d\sigma_0^{\text{vert}} & \text{diag}(h_1(p_0)I_R, \dots, h_N(p_0)I_R) \end{pmatrix}.$$

Its determinant is $\prod_{j=1}^N h_j(p_0)^R \neq 0$. If $b = 0$, this is just the lower right block.

The map from the pullback to $\text{Gen}_N(\mathcal{F}/B)$ is locally biholomorphic at p_0 , since $d\varphi_0$ is an isomorphism. Composing with it, and undoing the fixed GL_N change of basis, proves strong dominability at the originally prescribed point. This proves Theorem 3.1.

The bound ((3.1)) includes the deletion condition when $b = 0$. For a curve, it is

$$N \geq \max\{r + 2, r + b + 1\}.$$

The generating-section construction specializes to maps into projective space.

Let X be a fixed smooth connected projective variety of dimension $k \geq 1$, and put

$$q = h^1(X, \mathcal{O}_X).$$

Fix a nonempty component $B = \text{Pic}^\alpha(X)$. Assume that every $L \in B$ is globally generated and satisfies

$$H^1(X, L) = 0. \quad (3.6)$$

Equivalently for the present argument, one may assume directly that the Poincaré direct image is a vector bundle, commutes with base change, and has surjective evaluation.

Theorem 3.2. *If*

$$m \geq k + \max\{1, q\}, \quad (3.7)$$

then the full mapping space

$$\mathrm{Hol}_\alpha(X, \mathbb{P}^m) = \{f : X \rightarrow \mathbb{P}^m : f^*\mathcal{O}(1) \in B\}$$

is smooth, strongly dominable, and Oka.

This includes maps whose image lies in a proper projective linear subspace, maps with ramification, and maps that factor through another variety.

Normalize a Poincaré line bundle $\mathcal{P}$ on $X \times B$ at a fixed point of X . By cohomology and base change [4, Tag 07VJ], ((3.6)) gives

$$E = p_*\mathcal{P}$$

as a locally free sheaf with fibre $H^0(X, L)$, and the evaluation map is surjective.

A morphism to $\mathbb{P}^m$ is a line bundle equipped with $m + 1$ generating sections, modulo their common nonzero scalar. Consequently

$$\mathrm{Hol}_\alpha(X, \mathbb{P}^m) \cong \mathrm{Gen}_{m+1}(\mathcal{P}/B)/\mathbb{C}^*, \quad (3.8)$$

an open subset of $\mathbb{P}(E^{\oplus(m+1)})$.

This is an isomorphism of the natural complex spaces. The functorial statement is [3, Proposition 2.1], under hypotheses (A1) and (A2) of that paper. Families of morphisms pull back $\mathcal{O}(1)$ and its generating sections; conversely a family of generating sections defines the morphism. Tensoring by a line bundle from the parameter scheme accounts for the projectivization.

In particular, the space in ((3.8)) is smooth and the quotient map is a holomorphic principal $\mathbb{C}^*$ -bundle.

The Picard component B is a complex torus of dimension q , hence Oka. Apply Theorem 3.1 with $r = 1$, $b = q$, and $N = m + 1$. Condition ((3.1)) is exactly ((3.7)). Thus the generating-tuple space is strongly dominable.

The scalar quotient in ((3.8)) is a submersion. Composing a pointwise dominating map with this quotient gives a full-rank entire map to the mapping space. Restrict its domain to a linear complement of the one-dimensional differential kernel. This supplies the entire map from a Euclidean space of exactly the target dimension required by strong dominability. Now apply the strong dominability criterion [17, Theorem 1.3].

In the rank-one construction, the redundancy step has the following geometric form. The image $f(X)$ has dimension at most $k < m$, so a projective change of coordinates can put every coordinate vertex outside $f(X)$. Deleting any one coordinate then leaves a base-point-free tuple. The bad remaining-coordinate loci are closed analytic by compactness of X , and the complete flows are precisely ((3.5)).

The horizontal incidence calculation specializes to

$$\dim Z = M + k + q - (m + 1) < M.$$

Thus the construction applies at every point of the mapping space, including linearly degenerate maps.

The hypotheses are nonvacuous for every fixed smooth projective X . Given a line bundle L_0 and an ample line bundle A , there is N_0 such that for all $N \geq N_0$ and all $P \in \mathrm{Pic}^0(X)$,

$$L_0 \otimes A^N \otimes P$$

has vanishing H^1 , is globally generated, and is very ample. This follows by relative Serre vanishing [4, Tag 01XO] on $X \times \mathrm{Pic}^0(X)$, with a uniform choice possible because the Picard variety is

projective. One can impose the length-two restriction criterion uniformly using the proper length-two Hilbert scheme. See also [3, Lemma 3.7].

For the uniform statement, let $J = \text{Pic}^0(X)$ and $H = X^{[2]}$, and let $\mathcal{Z} \subset X \times H$ be the universal length-two subscheme. On $X \times J \times H$, pull back $\mathcal{P}$, L_0 , A , and $\mathcal{I}_{\mathcal{Z}}$ from their indicated factors. For the projection $q : X \times J \times H \rightarrow J \times H$, relative Serre vanishing [4, Tag 01XO] gives one N_0 such that

$$R^i q_*(L_0 \otimes \mathcal{P} \otimes A^N \otimes \mathcal{I}_{\mathcal{Z}}) = 0 \quad (i > 0, N \geq N_0),$$

and the same vanishing holds on every fibre. The fibrewise statement is uniform because the sheaf is flat over $J \times H$ and relative Serre vanishing applies after every base change. Flatness of $\mathcal{I}_{\mathcal{Z}}$ follows from the local flatness criterion [4, Tag 00MK] applied to

$$0 \longrightarrow \mathcal{I}_{\mathcal{Z}} \longrightarrow \mathcal{O}_{X \times H} \longrightarrow \mathcal{O}_{\mathcal{Z}} \longrightarrow 0,$$

whose middle and right terms are flat over H . Increase N_0 , if necessary, to obtain $H^1(X, L_0 \otimes A^N \otimes P) = 0$ for all $P \in J$ as well. The sequence

$$0 \longrightarrow (L_0 \otimes A^N \otimes P) \otimes \mathcal{I}_{\mathcal{Z}} \longrightarrow L_0 \otimes A^N \otimes P \longrightarrow (L_0 \otimes A^N \otimes P)|_{\mathcal{Z}} \longrightarrow 0$$

then gives surjectivity of restriction to every length-two Z . This is exactly the very-ampleness criterion used in Section 3.2 and also implies global generation.

For $m > k$, a general $(m+1)$ -tuple of sections of a globally generated line bundle has no common zero: the fixed-line-bundle incidence has codimension $m+1-k > 0$ in the parameter space. Thus the mapping spaces in the theorem are nonempty in these high-degree cases.

The construction applies to projective X of arbitrary Kodaira dimension. When $q = 0$, the conclusion is already accessible through the classical codimension-two algebraic-complement theorem; the construction above is especially useful for a moving Picard base.

Let C be any smooth projective curve of genus $g \geq 0$. If

$$d \geq 2g, \quad m \geq \max\{2, g+1\}, \quad (3.9)$$

then

$$\text{Hol}_d(C, \mathbb{P}^m)$$

is Oka. Indeed, every degree- d line bundle is globally generated and has $H^1 = 0$, because

$$H^1(C, L(-x)) \cong H^0(C, K_C \otimes L^{-1}(x))^* = 0.$$

Here $\deg(K_C \otimes L^{-1}(x)) = 2g - 1 - d < 0$.

Writing $R = d - g + 1$, the dimension is

$$g + (m+1)R - 1.$$

The theorem concerns the entire degree component for the fixed curve.

3.2 Embeddings and maps to Grassmannians

Keep the notation and hypotheses of Section 3.1, and suppose additionally that every $L \in B$ is very ample.

Theorem 3.3. *If*

$$m \geq 2k + \max\{2, q+1\}, \quad (3.10)$$

then the space

$$\text{Emb}_{\alpha}(X, \mathbb{P}^m)$$

of parametrized closed embeddings is smooth, strongly dominable, and Oka.

For a fixed curve C of genus g , sufficient numerical conditions are

$$d \geq 2g + 1, \quad m \geq \begin{cases} 4, & g = 0, \\ g + 3, & g \geq 1. \end{cases} \quad (3.11)$$

For a smooth k -fold X , the length-two Hilbert scheme $X^{[2]}$ is smooth projective of dimension $2k$. One construction is the quotient of $\mathrm{Bl}_\Delta(X \times X)$ by interchange of the two factors [6, §2.1]. Locally, use midpoint and difference coordinates; after blowing up the difference coordinates, the involution changes the sign of the one radial coordinate and fixes the remaining coordinates. Its quotient replaces that radial coordinate by its square and is smooth. The exceptional points record a point of X and a tangent direction.

For a line bundle L , a tuple $s_0, \dots, s_m$ defines a closed embedding precisely when

$$\mathbb{C}^{m+1} \longrightarrow H^0(Z, L|_Z)$$

is surjective for every $Z \in X^{[2]}$. The reduced Z test separation of distinct points; the nonreduced Z test separation of tangent directions. The same condition implies generation at every point. A proper injective immersion of a smooth variety is a closed embedding.

The target spaces $H^0(Z, L|_Z)$ form a rank-two tautological vector bundle on $X^{[2]}$. Very ampleness ensures that evaluation from $H^0(X, L)$ to each such target is surjective.

It follows that failure to embed is the proper image of the rank-at-most-one incidence over $X^{[2]}$. Hence it is a closed analytic condition in the section-parameter space. The embedding space is therefore an open subset of the smooth space in ((3.8)).

Pull the Poincaré direct image to $\mathbb{C}^q$ and trivialize it as in Section 3.1. Fix a tuple representing any prescribed embedding f_0 . Use the same finite-parameter family ((3.3)), with $N = m + 1$.

At $u \neq 0$ and $Z \in X^{[2]}$, the derivative in the coefficient parameters surjects onto

$$\mathrm{Mat}_{2 \times (m+1)}(\mathbb{C}).$$

The rank-at-most-one variety has codimension

$$(2 - 1)((m + 1) - 1) = m.$$

Separate its rank-zero and rank-one strata. Their inverse images are smooth because of parameter surjectivity, and their dimensions are at most

$$M + 2k + q - m < M$$

by ((3.10)). Applying the countable-atlas Sard argument from Section 3.1 to both strata produces one coefficient parameter for which every $u \neq 0$ gives an embedding. At $u = 0$, the given embedding is unchanged.

Thus an entire horizontal section through the prescribed embedding exists. If $q = 0$, it is the constant section at that point.

Let

$$\mathcal{S} = \{(Z, v) \in X^{[2]} \times \mathbb{P}^m : v \in \langle f_0(Z) \rangle\}.$$

The rank-two evaluation quotient makes $\mathcal{S} \rightarrow X^{[2]}$ a projective-line bundle. Hence $\mathcal{S}$ is projective of dimension $2k + 1$, and its image in $\mathbb{P}^m$ is a closed subset of dimension at most $2k + 1$. Reduced length-two schemes give secant lines and nonreduced ones give tangent lines, so this image contains exactly the closed union required for projection. By $m \geq 2k + 2$, it is proper. A basis whose projective vertices all avoid it is chosen successively, avoiding this proper closed set and the span of the previously chosen basis vectors.

Projection away from any coordinate vertex remains a closed embedding on $f_0(X)$: avoidance of the image makes the projection defined; avoidance of secants makes it injective; avoidance of tangent lines makes it immersive; properness makes it a closed embedding. Equivalently, deletion of any one section from the tuple still gives an embedding.

For each j , let A_j be the bad set where the other m sections fail the length-two surjectivity test. By Section 3.2 it is closed analytic in the corresponding Stein parameter space and does not contain the prescribed point. Choose h_j vanishing on A_j and nonzero at that point.

The flows

$$s_j \longmapsto s_j + t h_j(u, s_{\widehat{j}}) e_a$$

are complete and preserve embedding tuples. If the other sections already embed, their evaluations on every length-two scheme have rank two, and adding an arbitrary extra coordinate cannot reduce that rank. If they fail to embed, the coefficient is zero.

These fields span every vertical tangent direction at the prescribed point. Their finite composition with the horizontal section has full differential, exactly as in Section 3.1. Descend by common scalar multiplication and restrict to a linear complement of the differential kernel. This proves Theorem 3.3.

For $g \geq 2$, $\text{Aut}(C)$ is finite. Its precomposition action on $\text{Emb}_d(C, \mathbb{P}^m)$ is free, since $f \circ a = f$ and injectivity of f force $a = \text{id}$. Consequently

$$\text{Emb}_d(C, \mathbb{P}^m) / \text{Aut}(C)$$

is a smooth Oka manifold under the bounds ((3.11)).

This quotient parametrizes embedded images isomorphic to the fixed abstract curve C .

A parallel construction treats stable-quotient maps to Grassmannians.

Let C be a fixed smooth projective curve of genus $g \geq 2$, let $r \geq 2$, and let

$$d > r(2g - 1). \quad (3.12)$$

Assume that

$$(r, g) \neq (2, 2) \quad \text{or} \quad d \text{ is odd.} \quad (3.13)$$

Use the quotient convention for $\text{Gr}(r, N)$, and denote its rank- r universal quotient by Q . Define

$$\mathcal{M}_{d,N}^{\text{st}} = \{f : C \rightarrow \text{Gr}(r, N) : f^*Q \text{ is stable of rank } r \text{ and degree } d\}.$$

For a fixed degree- d line bundle ξ , define

$$\mathcal{M}_{\xi,N}^{\text{st}} = \{f \in \mathcal{M}_{d,N}^{\text{st}} : \det(f^*Q) \cong \xi\}.$$

The determinant datum is its isomorphism class ξ .

Theorem 3.4. *Under ((3.12))–((3.13)), these are smooth Oka manifolds, with the following sufficient bounds:*

<i>space</i>	<i>sufficient bound</i>	
$\mathcal{M}_{d,N}^{\text{st}}$	$N \geq r + r^2(g - 1) + 2$	
$\mathcal{M}_{\xi,N}^{\text{st}}$	$N \geq r + (r^2 - 1)(g - 1) + 1.$	

(3.14)

Theorem 2.1 supplies the Oka property of

$$U_C^s(r, d), \quad SU_C^s(r, \xi)$$

.

For a stable bundle E of rank r and degree d , Serre duality gives

$$H^1(C, E(-x))^* \cong \text{Hom}(E(-x), K_C).$$

A nonzero map would give a rank-one torsion-free quotient of $E(-x)$ whose degree is at most $2g - 2$. Stability forces the slope of a proper positive-rank quotient to be strictly greater than

$$\mu(E(-x)) = d/r - 1 > 2g - 2,$$

a contradiction. Thus $H^1(E(-x)) = 0$ for every x . The exact sequence for evaluation at x then proves global generation. The same slope argument gives $H^1(E) = 0$.

Consequently

$$R = h^0(C, E) = d + r(1 - g) \quad (3.15)$$

is constant throughout the stable base, and every genuine family of these bundles has locally free direct image, commuting with base change and with surjective evaluation.

A map $f : C \rightarrow \text{Gr}(r, N)$ is equivalent to a quotient

$$\mathcal{O}_C^N \twoheadrightarrow E$$

of rank r . This is the defining functor of the Grassmannian; see [4, Tag 089R].

For fixed stable E , quotient presentations are generating N -tuples in $H^0(C, E)$ modulo $\text{Aut}(E) = \mathbb{C}^*$. Thus the fibre of the forgetful map to the stable-bundle moduli space is

$$\text{Gen}_N(E)/\mathbb{C}^*. \quad (3.16)$$

The coarse-moduli base records the isomorphism class of the vector bundle, and the fibre quotient is by $\text{Aut}(E) = \mathbb{C}^*$.

On an analytic open set of that base carrying a universal family $\mathcal{E}$, ((3.16)) gives an open subset of

$$\mathbb{P}((p_*\mathcal{E})^{\oplus N}).$$

In particular the total mapping space is locally smooth, and the forgetful map is a holomorphic submersion. Its vertical tangent space at a tuple s is

$$H^0(C, E)^{\oplus N}/\mathbb{C}s. \quad (3.17)$$

The existence of these analytic local universal families, and of sufficient global families over the Euclidean dominating sources, is the next step.

Set $B = SU_C^s(r, \xi)$ and fix $y \in C$. There is a universal projective bundle, and its restriction at y is a holomorphic $\mathbb{P}^{r-1}$ -bundle

$$p_y : P_y \longrightarrow B.$$

There is a genuine universal vector bundle

$$\mathcal{U}_y \longrightarrow C \times P_y \quad (3.18)$$

whose fibre at a point of P_y represents the stable bundle specified by p_y .

An explicit primary source is [9, §2.1, equations (2.4) and (2.5)]. Their construction regards P_y as a parabolic moduli space with one multiplicity-one parabolic weight; the universal parabolic bundle supplies ((3.18)) after forgetting the flag. We use the smooth-curve specialization, in the genus and rank range stated in ((3.13)).

Analytic local triviality of P_y gives local holomorphic sections, and restriction of ((3.18)) along such a section gives the local universal family used in Section 3.2.

More importantly, let

$$\varphi : \mathbb{C}^b \rightarrow B, \quad \varphi(0) = b_0, \quad d\varphi_0 \text{ an isomorphism}$$

be an Oka dominating map, where $b = (r^2 - 1)(g - 1)$. The pullback φ^*P_y has structure group $\mathrm{PGL}_r(\mathbb{C})$. Since $\mathbb{C}^b$ is contractible Stein, the Oka–Grauert principle [27][20, Theorem 5.3.1] trivializes this principal group bundle holomorphically. Choose an entire section through any desired point of the fibre over zero. It gives

$$\tilde{\varphi} : \mathbb{C}^b \rightarrow P_y, \quad p_y \tilde{\varphi} = \varphi.$$

Pulling back ((3.18)) yields a genuine bundle

$$\mathcal{F} \longrightarrow C \times \mathbb{C}^b$$

whose classifying map is exactly φ .

At a prescribed point of $\mathcal{M}_{\xi, N}^{\mathrm{st}}$, choose an isomorphism between its quotient bundle and the fibre of the family just constructed at zero. By Section 3.2, the direct image of this family has rank ((3.15)) and has surjective evaluation.

Apply the explicit construction Section 3.1–Section 3.1 to this family over $\mathbb{C}^b$. The horizontal incidence has maximal dimension

$$M + 1 + b - (N - r + 1) = M + b + r - N.$$

It is less than M when $N \geq r + b + 1$. The deletion condition requires $N \geq r + 2$, which is automatic here since $b \geq 3$.

The resulting entire map into the generation space, followed by the map to $\mathcal{M}_{\xi, N}^{\mathrm{st}}$, has differential surjective on the stable-bundle base because $d\varphi_0$ is an isomorphism. Its complete vertical flows cover ((3.17)). Hence the full differential is surjective. Restrict to a linear complement of its kernel and apply the strong dominability criterion [17, Theorem 1.3]. This proves the second line of ((3.14)).

Let $J = \mathrm{Pic}^0(C)$. There is a finite étale map

$$SU_C^s(r, \xi) \times J \longrightarrow U_C^s(r, d), \quad (E, L) \longmapsto E \otimes L. \quad (3.19)$$

More precisely, identify $\mathrm{Pic}^d(C)$ with J by tensoring with ξ^{-1} . Then ((3.19)) is the base change of

$$[r] : J \rightarrow J, \quad L \mapsto L^{\otimes r},$$

along the determinant map. The deck group $J[r]$ acts by

$$T \cdot (E, L) = (E \otimes T, L \otimes T^{-1}).$$

Translation on the second factor is free for $T \neq \mathcal{O}_C$, so ((3.19)) is unramified even when E admits a self-twist.

The covering base is Oka. On a Euclidean dominating source for this product, pull back the family obtained from P_y on the first factor and tensor it with the Poincaré line bundle from the second factor. This is a genuine family whose classifying map to $U_C^s(r, d)$ has full differential. Repeat Section 3.2 with

$$b = r^2(g - 1) + 1.$$

The first line of ((3.14)) follows.

The variable-determinant dimension is

$$\begin{aligned} \dim \mathcal{M}_{d, N}^{\mathrm{st}} &= r^2(g - 1) + 1 + N(d + r(1 - g)) - 1 \\ &= Nd + r(N - r)(1 - g). \end{aligned}$$

Fixing the determinant subtracts g . This also matches the expected tangent-space calculation: from

$$0 \rightarrow \text{End}(E) \rightarrow E^{\oplus N} \rightarrow f^*T \text{Gr}(r, N) \rightarrow 0$$

and $H^1(E) = 0$, one obtains $H^1(f^*T \text{Gr}(r, N)) = 0$.

For example, for any fixed genus-two curve and stable rank-two quotient of degree seven, $\mathcal{M}_{7,8}^{\text{st}}$ is Oka of dimension 44. With determinant fixed, $\mathcal{M}_{\xi,6}^{\text{st}}$ is Oka of dimension 32.

4 Hilbert and Quot schemes

Hilbert and Quot schemes enter through Grothendieck's representability constructions; the Hilbert-scheme construction is presented in [28]. The arguments below use the resulting moduli spaces together with intrinsic strata adapted to the Oka union theorem.

For a complex manifold S , write

$$F_r(S) = \{(p_1, \dots, p_r) \in S^r : p_i \neq p_j \text{ for } i \neq j\}.$$

Theorem 4.1. *Let S be a smooth quasiprojective complex surface. If $F_r(S)$ is Oka for every $1 \leq r \leq n$, then $S^{[n]}$ is Oka.*

The proof uses intrinsic Hilbert–Samuel strata, which are preserved by changes of local coordinates.

Let $A = \mathbb{C}[[u, v]]$ and $\mathfrak{m} = (u, v)$. For a colength- m ideal I , put

$$\chi_I(j) = \dim_{\mathbb{C}} A/(I + \mathfrak{m}^{j+1}), \quad 0 \leq j \leq m-1.$$

These cumulative values are intrinsic. Let Z_T be the reduced locally closed punctual locus on which all these values have a fixed type T .

The geometry of punctual Hilbert schemes and their Hilbert–Samuel strata was developed by Briançon [12] and Iarrobino [34]. We use the following statements in the precise modern form proved by Bayer–Chen–Jiang:

1. Every nonempty Z_T is smooth [5, Theorem 2.4].
2. The normal-pattern chart $Z_P \subset Z_T$ is a Zariski open affine space [5, Definition 2.8 and Corollary 2.14].
3. In characteristic zero, finitely many GL_2 -translates of Z_P cover Z_T [5, Proposition 2.10].

Thus every entire intrinsic stratum Z_T is Oka by Zariski localization. For $m = 1$, the punctual Hilbert scheme is a point.

The filtration below is obtained from upper semicontinuity of the Hilbert–Samuel profile, independently of any frontier property for individual strata; compare [5, discussion following Theorem 2.4].

A local algebra of length at most n satisfies $\mathfrak{m}^n = 0$, so every relevant ideal contains $\mathfrak{m}^n$. Indeed, in the Artinian quotient $Q = A/I$ let $\mathfrak{n} = \mathfrak{m}Q$. If $\mathfrak{n}^j = \mathfrak{n}^{j+1}$, Nakayama's lemma applied to the finite Q -module $\mathfrak{n}^j$ gives $\mathfrak{n}^j = 0$. Therefore every nonzero term in

$$Q \supset \mathfrak{n} \supset \mathfrak{n}^2 \supset \dots$$

is followed by a strict decrease in vector-space dimension. Since $\dim Q \leq n$ and $\dim(Q/\mathfrak{n}) = 1$, the n th power is zero. Truncated local coordinates therefore suffice.

Fix an ordered length list $(m_1, \dots, m_r)$ with sum n , and intrinsic types $(T_1, \dots, T_r)$. Over $F_r(S)$, the corresponding ordered Hilbert locus is a holomorphic locally trivial bundle with fibre

$$Z_{T_1} \times \dots \times Z_{T_r}. \tag{4.1}$$

Indeed, local coordinates at the moving support points identify their truncated local rings with $\mathbb{C}[u, v]/(u, v)^n$. The transition maps belong to the finite-dimensional group of automorphisms of this local algebra. They preserve the maximal-ideal filtration and hence preserve every Z_T . The locus is the associated bundle for the corresponding product of coordinate-jet principal bundles.

The base and fibre in ((4.1)) are Oka. The ordered total space is therefore Oka and smooth. To forget the ordering, divide by the finite group of permutations preserving the length and type labels. Its action is free because the support points are distinct. The resulting unlabelled type locus is again smooth and Oka by unramified-cover invariance.

First filter the Hilbert scheme by the number of distinct support points. Under Hilbert–Chow,

$$S^{[n]} \longrightarrow \mathrm{Sym}^n S,$$

the condition “support has at most r points” is closed: in S^n it is a finite union of diagonals, whose image under the finite symmetric-group quotient is closed. Its inverse image in the Hilbert scheme is therefore closed.

On the stratum with exactly r support points, the multiplicity partition is locally constant. This can be checked in disjoint small neighborhoods of the support points: while the number of support points remains r , each neighborhood contains exactly one support point, with its unchanged cycle multiplicity.

Fix such a partition and pass to the finite étale cover ordering its support points. For each point i and each $0 \leq j < n$, the integer

$$\chi_i(j) = \dim \mathcal{O}_{S, p_i} / (I_i + \mathfrak{m}_{p_i}^{j+1})$$

is upper semicontinuous. It is the fibre dimension of a coherent cokernel, so the sets $\{\chi_i(j) \geq a\}$ are closed by the corresponding Fitting-ideal condition [4, Tags 07Z6 and 0C3C].

Define the symmetric integer function

$$c = \sum_{i=1}^r \sum_{j=0}^{n-1} \chi_i(j). \quad (4.2)$$

It is upper semicontinuous and invariant under permutations, and hence descends to the unlabelled multiplicity stratum. Its finitely many superlevel sets give a finite relative closed filtration.

On a fixed level $c = c_0$, the full ordered Hilbert–Samuel profile is locally constant. To see this, near a given profile each of the finitely many upper-semicontinuous integer functions is at most its value at that point. If their sum is unchanged, none can decrease. Thus all are locally constant on the level.

Consequently each difference in the filtration by c is a finite disjoint union of the smooth Oka loci constructed in Section 4. Profiles sharing the same value of c form separate open-and-closed pieces of that level.

For an explicit global indexing, put

$$H_r = \{Z \in S^{[n]} : \# \mathrm{Supp} Z \leq r\}, \quad H_0 = \emptyset.$$

On $H_r \setminus H_{r-1}$ the multiplicity partitions are open and closed, so their functions ((4.2)) combine into a single upper-semicontinuous function c . It satisfies $0 \leq c \leq n^2$, because $\chi_i(j) \leq m_i$ and $\sum_i m_i = n$. Define

$$F_{r,a} = H_{r-1} \cup \{Z \in H_r \setminus H_{r-1} : c(Z) \geq a\}, \quad 0 \leq a \leq n^2 + 1.$$

The complement of $F_{r,a}$ in H_r is $\{c < a\}$, open in the open set $H_r \setminus H_{r-1}$; hence $F_{r,a}$ is closed in H_r and in $S^{[n]}$. Moreover,

$$H_{r-1} = F_{r,n^2+1} \subset F_{r,n^2} \subset \cdots \subset F_{r,0} = H_r.$$

Concatenate these chains for $r = 1, \dots, n$. Each successive difference is a fixed level of c , and hence the finite disjoint union of smooth Oka type loci already proved above. This is the required finite closed filtration.

Finally, $S^{[n]}$ is smooth by Fogarty's theorem [18, Theorem 2.4]. Every difference is smooth Oka, so the finite Oka stratification theorem [17, Corollary 4.5] proves Theorem 4.1.

We next turn from Hilbert schemes to nested Quot schemes on curves. Their smoothness and motivic geometry were studied by Monavari–Ricolfi [41]; the filtration below records the exact ranks needed for the Oka deduction.

Theorem 4.2. *Let C be a smooth connected projective curve of genus at most one, let V be any rank- r vector bundle, and let $0 \leq d_1 \leq \dots \leq d_s$. The projective scheme parametrizing*

$$K_s \subset \dots \subset K_1 \subset V, \quad \text{length}(V/K_j) = d_j,$$

is smooth and Oka. Its dimension is rd_s .

Proof: smoothness.

A submodule of a free module over the local discrete valuation ring of a smooth curve is free. Hence every K_j is locally free. Choose the quotients successively: first V/K_1 , then K_1/K_2 , and so on. At a step whose kernel is K and zero-dimensional quotient is T , the obstruction group is

$$\text{Ext}^1(K, T) = H^1(C, K^* \otimes T) = 0.$$

The last equality holds because a sheaf supported on finitely many points has no positive cohomology. The relative Quot morphism is therefore smooth. Its relative tangent space is $\text{Hom}(K, T)$, of dimension $r \text{length}(T)$. Summing the increments gives rd_s . Each step is projective by the Quot construction. The universal kernels are locally free: they are flat over the base and their fibres are locally free of constant rank, so the local flatness criterion [4, Tag 00MK] and Nakayama's lemma apply.

Proof: rank one.

For a line bundle L , every kernel is $L(-D_j)$ for a unique effective divisor, and nesting is $D_1 \leq \dots \leq D_s$. Taking successive differences gives

$$\text{Quot}_{\mathbf{d}}(L) \cong \prod_{j=1}^s \text{Sym}^{d_j - d_{j-1}} C, \quad d_0 = 0.$$

For $C = \mathbb{P}^1$, $\text{Sym}^a C = \mathbb{P}^a$. For an elliptic curve and $a \geq 1$, Abel–Jacobi is a projective-space bundle of fibre $\mathbb{P}^{a-1}$ over $\text{Pic}^a C$: every degree- a line bundle has $H^1 = 0$ and $h^0 = a$, so its spaces of sections form a vector bundle. For $a = 0$, the symmetric power is a point. These spaces are Oka.

Proof: one rank-reduction step.

Choose a line subbundle $L \subset V$ with locally free quotient W . Such a subbundle exists: a nonzero section of a sufficiently positive twist gives a rank-one subsheaf, and saturation makes its quotient torsion-free and therefore locally free. Fix a kernel $K \subset V$, and put

$$L' = K \cap L = L(-D), \quad W' = \text{im}(K \rightarrow W).$$

Both L' and W' are locally free. If $a = \deg D$ and $b = \text{length}(W/W')$, the exact sequence for V/K gives $a + b = d$. Let $E' \subset V$ be the inverse image of W' . The required K corresponds bijectively to a splitting of

$$0 \rightarrow L/L' \rightarrow E'/L' \rightarrow W' \rightarrow 0.$$

Indeed, the image of a splitting pulls back to a submodule $K \subset E'$ meeting L in L' ; conversely, $K/L' \rightarrow W'$ is an isomorphism and gives the splitting. The extension splits since

$$\mathrm{Ext}^1(W', L/L') = H^1(C, (W')^* \otimes L/L') = 0.$$

Its splittings form a torsor under $\mathrm{Hom}(W', L/L')$, of dimension $(r-1)a$.

Here is the family calculation. Over a parameter scheme B carrying $L' \subset L$ and $W' \subset W$, let $\pi : C \times B \rightarrow B$, and write

$$\mathcal{T} = L/L', \quad \mathcal{H} = \mathcal{H}om(W', \mathcal{T}) = (W')^\vee \otimes \mathcal{T}.$$

The sheaf $\mathcal{T}$ is flat of relative length a and has support finite over B . Locally along that support W' is free of rank $r-1$. Therefore $\mathcal{H}$ is B -flat of relative length $(r-1)a$, and

$$\pi_* \mathcal{H} \text{ is locally free of rank } (r-1)a, \quad R^q \pi_* \mathcal{H} = 0 \quad (q > 0).$$

Finite pushforward commutes with base change. If B is affine, the Leray sequence gives

$$H^1(C \times B, \mathcal{H}) = H^1(B, \pi_* \mathcal{H}) = 0.$$

Since W' is locally free, this is the group $\mathrm{Ext}_{C \times B}^1(W', \mathcal{T})$ classifying the displayed extension. It vanishes, so the extension splits on each affine open of B . Two splittings differ by a section of $\pi_* \mathcal{H}$. Hence the parameter space of splittings is a genuine Zariski locally trivial affine-space torsor of the stated rank.

Proof: compatibility of a flag.

Set $L'_j = K_j \cap L = L(-D_j)$ and $W'_j = \mathrm{im}(K_j \rightarrow W)$, and write

$$a_j = \deg D_j, \quad b_j = d_j - a_j.$$

Both sequences are nondecreasing. Fix the divisors and the flag $W'_s \subset \cdots \subset W'_1 \subset W$. After K_{j-1} has been chosen, let $E_j \subset K_{j-1}$ be the inverse image of W'_j . Choosing K_j is equivalent to splitting

$$0 \rightarrow L'_{j-1}/L'_j \rightarrow E_j/L'_j \rightarrow W'_j \rightarrow 0.$$

The obstruction group vanishes by the same finite-support computation. The choices form an affine torsor of rank $(r-1)(a_j - a_{j-1})$, with $a_0 = 0$. Hence the exact-**a** stratum is an affine-bundle tower over

$$\left(\prod_j \mathrm{Sym}^{a_j - a_{j-1}} C \right) \times \mathrm{Quot}_{\mathbf{b}}(W).$$

By induction on r , this base is Oka; the successive affine bundles are Oka.

Proof: closed filtration and conclusion.

The number a_j is the length of the image of $L \rightarrow V/K_j$. Let $\mathcal{Q}$ be the full projective nested-Quot scheme, with universal kernels $\mathcal{K}_j$, and set

$$\mathcal{L}_j = \ker(p_C^* L \rightarrow p_C^* V/\mathcal{K}_j).$$

These sheaves are coherent before imposing any rank stratum. Choose one positive twist H^m so large that

$$R^1 p_*(\mathcal{L}_j \otimes p_C^* H^m) = 0 \quad \text{for every } j, \quad \deg(L \otimes H^m) - d_s > 2g - 2.$$

The first conditions follow simultaneously from relative Serre vanishing [4, Tag 01XO]; the last gives $H^1(L(-D) \otimes H^m) = 0$ for every divisor of degree at most d_s . Exact pushforward of the

universal image sequence identifies the sheaf image below with the pushforward of the universal finite sheaf image:

$$H^0(C, L \otimes H^m) \otimes \mathcal{O} \longrightarrow p_*((V/K_j) \otimes H^m).$$

The target is a vector bundle of rank d_j . The condition $a_j \leq a$ is therefore closed, being a determinantal rank condition. There are finitely many possible vectors $\mathbf{a}$. Order them by a linear extension of the componentwise partial order, with smaller vectors first. Every initial set is downward closed and is a finite union of sets of the form

$$\bigcap_j \{a_j \leq e_j\}.$$

It is therefore closed. The corresponding successive differences have one exact rank vector.

We now identify these differences in families. Near a fixed parameter, choose an affine open of C containing the finite support of every quotient in that fibre; properness of the support morphism lets us shrink the parameter neighbourhood so that all nearby supports stay in that affine open. Finite pushforward on the resulting supports is exact. A matrix of constant rank has locally free image and cokernel: locally invert a maximal nonzero minor and use row and column operations to reduce the matrix to $\text{diag}(I_a, 0)$. Applied to the displayed evaluation maps, this proves flatness of the finite image of L and its quotient. The prescribed relative Serre vanishing makes the evaluation image equal to the pushforward of the universal sheaf image before restriction to a stratum; the fibrewise vanishing identifies its rank with a_j . Thus $L/(K_j \cap L)$ is flat of length a_j , and the corresponding ideal in L is a relative effective Cartier divisor D_j . The quotient W/W'_j is flat of length $b_j = d_j - a_j$ by the exact sequence

$$0 \rightarrow L/(K_j \cap L) \rightarrow V/K_j \rightarrow W/W'_j \rightarrow 0.$$

Its kernel W'_j is a vector bundle by the local flatness criterion [4, Tag 00MK] already used in the smoothness proof. These constructions commute with base change and give the divisor and nested-Quot morphisms. Conversely, the successive splitting construction recovers precisely such kernels and flags. It is inverse on families, so each exact-rank difference is the smooth Oka affine-bundle tower described above, with the scheme structure described above. Apply the finite Oka stratification theorem [17, Corollary 4.5]. $\square$

The affine torsors and the closed filtration used here are constructed above for the given bundle itself.

5 Character varieties and parabolic connections

5.1 Character and representation varieties

Let Σ_g be a closed oriented surface of genus $g \geq 2$ and write

$$\mathcal{X}_g = \text{Hom}(\pi_1 \Sigma_g, \text{SL}_2(\mathbb{C})) // \text{SL}_2(\mathbb{C}).$$

Its regular locus is the irreducible-character locus. Goldman's symplectic construction makes this locus a holomorphic symplectic manifold of dimension $6g - 6$ [23].

Proposition 5.1. *The irreducible-character locus $\mathcal{X}_g^{\text{irr}}$ is holomorphically elliptic and hence Oka.*

Proof. For a simple closed curve γ , let

$$f_\gamma([\rho]) = \text{tr } \rho(\gamma).$$

With the trace pairing, the variation of the trace is represented by

$$F(\rho(\gamma)) = \rho(\gamma) - \frac{1}{2} \text{tr}(\rho(\gamma))I.$$

The complex twist flow is defined using

$$\zeta_t = \exp(tF(\rho(\gamma))), \quad t \in \mathbb{C}. \quad (5.1)$$

For a nonseparating curve, cut the surface and multiply the appropriate HNN stable letter by ζ_t . For a separating curve, conjugate the representation on one side by ζ_t . The matrix ζ_t commutes with $\rho(\gamma)$, so the gluing relation is preserved.

The construction is holomorphic for every complex time. Along the twist, the matrix associated with γ is unchanged, so replacing t by $-t$ gives the inverse. It also preserves irreducibility. In the separating case, an invariant line for the twisted representation is invariant under $\rho(\gamma)$ and hence under ζ_t ; it is therefore invariant under the original representation on both sides. The HNN argument is identical after observing that invariance under the cut-surface group implies invariance under ζ_t .

These twist flows were introduced as Hamiltonian flows by Goldman [24]. For the complex SL_2 formula used here, see [25, §2.3, equation (2.5), and Lemma 2.3.1].

For $A, B \in \mathrm{SL}_2$,

$$\mathrm{tr}(AB) + \mathrm{tr}(AB^{-1}) = \mathrm{tr}(A)\mathrm{tr}(B), \quad (5.2)$$

by Cayley–Hamilton applied to B . Resolving crossings of immersed curves with ((5.2)) expresses trace functions of arbitrary loops as polynomials in traces of simple closed curves. The trace-algebra statement and its skein formulation are treated in [46, Fact 4.1 and Theorem 7.1].

The coordinate ring of the affine SL_2 character variety is generated by traces and is finitely generated. Therefore finitely many simple traces $f_1, \dots, f_N$ suffice: first choose finitely many trace-polynomial generators and then take the finite union of the simple traces occurring in their crossing resolutions.

The morphism

$$(f_1, \dots, f_N) : \mathcal{X}_g \hookrightarrow \mathbb{C}^N$$

is a closed embedding, so $df_1, \dots, df_N$ span the cotangent space at every smooth point. The symplectic form identifies these differentials with Hamiltonian vector fields. By Section 5.1 each field is complete.

Their finite flow composition is a global dominating spray on $\mathcal{X}_g^{\mathrm{irr}}$. Thus this locus is holomorphically elliptic and hence Oka.

In detail, if W_i is the Hamiltonian field of f_i and Ψ_i^t is its complete flow, define

$$\mathfrak{s}(x, t_1, \dots, t_N) = \Psi_N^{t_N} \circ \dots \circ \Psi_1^{t_1}(x).$$

Then $\mathfrak{s}(x, 0) = x$ and

$$d\mathfrak{s}_{(x,0)}(0, \tau_1, \dots, \tau_N) = \sum_{i=1}^N \tau_i W_i(x).$$

Nondegeneracy of the symplectic form and spanning of the $df_i(x)$ make this differential surjective. Thus $\mathfrak{s}$ is a dominating spray on the trivial vector bundle $\mathcal{X}_g^{\mathrm{irr}} \times \mathbb{C}^N$.

The identification of the smooth and irreducible loci is also recorded in [7, Lemma 2.4]. $\square$

Proposition 5.2. *For an oriented surface with punctures or boundary and fixed regular semisimple $\mathrm{SL}_2(\mathbb{C})$ boundary conjugacy classes, the irreducible relative-character locus is holomorphically elliptic and hence Oka.*

Proof. The same proof applies to an oriented surface with punctures or boundary, with fixed regular semisimple SL_2 boundary conjugacy classes. The irreducible relative-character locus is a smooth Goldman symplectic leaf. Twists along interior simple curves preserve the prescribed boundary conjugacy classes.

Simple trace functions generate the relative coordinate ring after the boundary traces are fixed. Restricting a finite generating set therefore spans the cotangent space of the smooth relative locus, and its Hamiltonian fields are complete by ((5.1)).

For the boundary version of the trace and twist descriptions, see [26, Theorem 2.1, Lemma 3.1, and §4]. The matrix-exponential calculation in Section 5.1 gives completeness for every complex time.

Consequently the irreducible relative-character loci used in Section 5.2 are also elliptic. $\square$

We next consider the smooth representation varieties themselves.

Let

$$R_g = \text{Hom}(\pi_1 \Sigma_g, \text{SL}_2(\mathbb{C})), \quad g \geq 2.$$

Theorem 5.3. *Both*

$$(R_g)_{\text{reg}} \quad \text{and} \quad \text{Hom}(\pi_1 \Sigma_g, \text{GL}_2(\mathbb{C}))_{\text{reg}}$$

are Oka.

A representation is called Schur if

$$\text{End}_{\pi_1 \Sigma_g}(\mathbb{C}^2) = \mathbb{C}I.$$

We first identify this locus with the full smooth locus, paying attention to reducedness.

[7, Theorem 2.1] records that the GL_n surface-group representation scheme is reduced and a complete intersection for $g > 1$. The finite étale determinant-square-root map

$$R_g \times (\mathbb{C}^*)^{2g} \longrightarrow \text{Hom}(\pi_1 \Sigma_g, \text{GL}_2)$$

therefore gives reducedness and pure dimension $6g - 3$ for R_g .

The scheme tangent space at ρ is

$$Z^1(\Sigma_g, \text{ad } \rho).$$

Write $h^0 = \dim H^0(\Sigma_g, \text{ad } \rho)$. The trace pairing makes the adjoint local system self-dual. Poincaré duality and the cellular Euler characteristic give

$$\dim H^2 = h^0, \quad \dim H^1 = 6g - 6 + 2h^0.$$

Since $\dim B^1 = 3 - h^0$,

$$\dim T_\rho R_g = 6g - 3 + h^0. \tag{5.3}$$

The numbers here follow from the cellular cochain complex of the standard cell decomposition with one zero-cell, $2g$ one-cells, and one two-cell:

$$C^0 = \mathfrak{sl}_2, \quad C^1 = \mathfrak{sl}_2^{2g}, \quad C^2 = \mathfrak{sl}_2.$$

Its Euler characteristic is $3 - 6g + 3 = 6 - 6g$. Therefore

$$h^0 - \dim H^1 + h^0 = 6 - 6g, \quad \dim Z^1 = \dim H^1 + \text{rank}(d^0) = \dim H^1 + 3 - h^0,$$

which gives ((5.3)). Infinitesimally, writing a deformation as $\rho_\varepsilon(\gamma) = (I + \varepsilon u(\gamma))\rho(\gamma)$ with $\varepsilon^2 = 0$ yields

$$u(\gamma\delta) = u(\gamma) + \text{Ad}_{\rho(\gamma)} u(\delta),$$

the defining cocycle equation. Thus the calculation takes place in the representation scheme.

Because the scheme is reduced and pure of dimension $6g - 3$, ((5.3)) shows that ρ is smooth exactly when $h^0 = 0$, equivalently exactly when it is Schur.

The reduced complete-intersection structure established above identifies the scheme tangent calculation with the tangent space of the reduced representation variety; see [52] for this distinction.

The irreducible locus R_g^{irr} is a holomorphic principal PGL_2 -bundle over $\mathcal{X}_g^{\text{irr}}$. The action is free after quotienting the scalar centre, and the stable quotient supplies analytic local sections. This bundle statement is also used in the proof of [7, Theorem 1.3].

The base is Oka by Section 5.1 and the fibre is a complex Lie group. Thus R_g^{irr} is Oka.

A reducible representation can be made upper triangular. Write its diagonal character as

$$\lambda = (\lambda_1, \mu_1, \dots, \lambda_g, \mu_g) \in (\mathbb{C}^*)^{2g}.$$

It is Schur precisely when the two diagonal characters λ, λ^{-1} are distinct and the extension is nonsplit.

To verify this, if $\lambda^2 \neq 1$, choose a generator with distinct eigenvalues and diagonalize it. Any commuting traceless matrix is diagonal. If the representation is nonsplit, some other matrix has a nonzero off-diagonal entry, forcing that diagonal traceless matrix to vanish. Conversely, a split representation has a nonzero diagonal traceless centralizer. If $\lambda^2 = 1$, every matrix is a scalar times an upper unipotent matrix and commutes with a nonzero nilpotent traceless matrix.

Thus the base of diagonal-character data is

$$B = (\mathbb{C}^*)^{2g} \setminus \{\pm 1\}^{2g},$$

which is Oka by Corollary 1.6.

In an upper-triangular frame, write

$$\rho(\gamma) = \begin{pmatrix} \lambda(\gamma) & b(\gamma)\lambda(\gamma)^{-1} \\ 0 & \lambda(\gamma)^{-1} \end{pmatrix}.$$

The homomorphism condition is

$$b(\gamma\delta) = b(\gamma) + \lambda(\gamma)^2 b(\delta).$$

For the standard generators a_i, b_i of the surface group, write the cocycle values as p_i, q_i . The sole surface relation becomes

$$\sum_{i=1}^g ((1 - \mu_i^2)p_i + (\lambda_i^2 - 1)q_i) = 0. \quad (5.4)$$

Indeed, the cocycle of each commutator is the displayed summand, and its diagonal character is one, so the cocycle of the product is their sum.

Put $\chi = \lambda^2$. The relation becomes

$$b(\gamma^{-1}) = -\chi(\gamma)^{-1}b(\gamma).$$

For $[a_i, b_i] = a_i b_i a_i^{-1} b_i^{-1}$, the cocycle identity gives

$$\begin{aligned} b([a_i, b_i]) &= p_i + \lambda_i^2 q_i - \lambda_i^2 \mu_i^2 \lambda_i^{-2} p_i - \mu_i^2 \mu_i^{-2} q_i \\ &= (1 - \mu_i^2)p_i + (\lambda_i^2 - 1)q_i. \end{aligned}$$

Since $\chi([a_i, b_i]) = 1$, the product of commutators gives precisely ((5.4)).

On B , the linear functional in ((5.4)) never vanishes identically. Its kernel is therefore a rank- $(2g - 1)$ vector bundle $\mathcal{Z}$. Split extensions form the line subbundle $\mathcal{B}$ generated by

$$(1 - \lambda_1^2, 1 - \mu_1^2, \dots, 1 - \lambda_g^2, 1 - \mu_g^2). \quad (5.5)$$

This is nowhere zero on B . Adapted local frames for the pair $(\mathcal{Z}, \mathcal{B})$ identify its complement with

$$\mathbb{C} \times (\mathbb{C}^{2g-2} \setminus \{0\}). \quad (5.6)$$

Let the unique invariant line vary in $\mathbb{P}^1$. Associate the data $\mathcal{Z} \setminus \mathcal{B}$ to the Borel principal bundle

$$\mathrm{SL}_2 \rightarrow \mathbb{P}^1.$$

Conjugation by upper triangular matrices preserves the diagonal character and the split-extension subbundle. Local triangular frames and adapted frames for $(\mathcal{Z}, \mathcal{B})$ give a holomorphic locally trivial bundle over

$$\mathbb{P}^1 \times B$$

with fibre ((5.6)).

In coordinates this action has the form

$$\begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix} \begin{pmatrix} \lambda & b\lambda^{-1} \\ 0 & \lambda^{-1} \end{pmatrix} \begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix}^{-1} = \begin{pmatrix} \lambda & (t^2b + tu(1 - \lambda^2))\lambda^{-1} \\ 0 & \lambda^{-1} \end{pmatrix}.$$

Thus the action on a cocycle vector is $b \mapsto t^2b + tu v_\lambda$, where v_λ is the nowhere-zero vector ((5.5)). It is a holomorphic affine automorphism of each fibre of $\mathcal{Z}$, preserves the line $\mathcal{B}$, and preserves its complement. On an open set of B choose a vector-bundle frame whose first vector is v_λ ; the complement becomes

$$\{(z_0, z_1, \dots, z_{2g-2}) : (z_1, \dots, z_{2g-2}) \neq 0\}.$$

Together with local sections of $\mathrm{SL}_2 \rightarrow \mathbb{P}^1$, this gives the claimed local trivializations explicitly.

Its total space is the reducible Schur locus in the matrix representation variety.

The reducible locus is closed in R_g : it is the proper image of the closed invariant-line incidence in $R_g \times \mathbb{P}^1$. Its intersection with the Schur locus is therefore closed.

A nonsplit reducible representation has a unique invariant line. The assignment of this line is holomorphic locally. Choose a generator with distinct eigenvalues; its two eigenlines vary holomorphically. At the chosen representation, one of them fails to be invariant under another generator, and this failure persists in a neighborhood. The remaining eigenline is consequently the unique common invariant line throughout the reducible neighborhood. This proves that the bundle in Section 5.1 has exactly the induced complex-manifold structure of the closed reducible Schur locus.

The base $\mathbb{P}^1 \times B$ is Oka, and the fibre ((5.6)) is Oka because $2g - 2 \geq 2$. Therefore this closed submanifold is Oka. Its complement is the Oka irreducible locus from Section 5.1. Applying the Oka union theorem [17, Theorem 1.5], and using Section 5.1, proves that $(R_g)_{\mathrm{reg}}$ is Oka.

Finally, the finite étale determinant-square-root quotient preserves smooth loci and gives the GL_2 statement. Its dimension is $8g - 3$, whereas the SL_2 smooth representation locus has dimension $6g - 3$.

5.2 Parabolic connections and Garnier spaces

Let C be a fixed smooth projective curve of genus g , let

$$D = t_1 + \dots + t_n$$

be a reduced marked divisor, and choose exponents

$$\lambda_i \in \mathbb{C}, \quad 2\lambda_i \notin \mathbb{Z}. \quad (5.7)$$

Consider rank-two logarithmic parabolic connections with trivial determinant and its trivial connection, residue eigenvalues $\lambda_i, -\lambda_i$, and the prescribed marked eigenline at each puncture. Take generic parabolic weights α , so stability equals semistability.

We use the existence-theorem range

$$g = 0, n \geq 4; \quad g = 1, n \geq 2; \quad g \geq 2, n \geq 1. \quad (5.8)$$

Denote the stable moduli space by

$$\mathcal{M}_{\text{dR}}^\alpha(C, D; \lambda).$$

Theorem 5.4. *Under ((5.7))–((5.8)), this entire smooth moduli space is Oka. Its dimension is*

$$6g - 6 + 2n.$$

For $C = \mathbb{P}^1$, the theorem gives the fixed-time initial-data moduli spaces of nonresonant Garnier systems with $n \geq 4$ punctures.

Set

$$\mu_i = e^{-2\pi i \lambda_i}.$$

Condition ((5.7)) makes $\mu_i \neq \mu_i^{-1}$. Let $\mathcal{X}$ be the relative SL_2 character variety with puncture traces $\mu_i + \mu_i^{-1}$.

The moduli existence, smoothness, and Riemann–Hilbert results used are:

- [35, Theorems 1.1, 1.3, 1.4, 1.5], for the genus-zero construction.
- [36, Theorems 2.1 and 2.2], for the general-curve construction; §5 gives smoothness of the determinant map, so fixing determinant and its connection gives the required smooth fibre.

The Riemann–Hilbert morphism

$$\text{RH} : \mathcal{M}_{\text{dR}}^\alpha \longrightarrow \mathcal{X}$$

is holomorphic, and over irreducible characters it is an analytic isomorphism. Its irreducible preimage is therefore Oka by Section 5.1.

Nonresonance also gives a family version of the logarithmic Riemann–Hilbert correspondence with these specified residue branches. The classical regular-singular extension is Deligne’s construction [15, Chapter II]. Locally the distinct eigenlines determine the lattice uniquely. Apply the Deligne extension and then the prescribed integer elementary transformations along those eigenlines. This is compatible with holomorphic families; see [36, Proposition 3.1 and §6]. This family correspondence identifies the exceptional part analytically.

The local construction can be made explicit. If T_i is the puncture monodromy matrix in a holomorphic family, set

$$P_i^+ = \frac{T_i - \mu_i^{-1}I}{\mu_i - \mu_i^{-1}}, \quad P_i^- = \frac{T_i - \mu_i I}{\mu_i^{-1} - \mu_i}, \quad A_i = \lambda_i P_i^+ - \lambda_i P_i^-.$$

The Cayley–Hamilton identity $(T_i - \mu_i I)(T_i - \mu_i^{-1}I) = 0$ gives

$$(P_i^\pm)^2 = P_i^\pm, \quad P_i^+ P_i^- = 0, \quad P_i^+ + P_i^- = I, \quad e^{-2\pi i A_i} = T_i.$$

Thus

$$d + A_i \frac{dz}{z}$$

is a holomorphic family of logarithmic connections on the disc, with exactly the required residue branches and local monodromy. Over the punctured disc, the flat isomorphism determined by a frame at one point identifies this model with the given local system. These identifications are holomorphic in the parameters. Gluing these disc models to the flat bundle on $C \setminus D$ constructs the required family.

Nonresonance gives uniqueness and holomorphic dependence at each puncture. After choosing holomorphic local eigenframes in the parameter space, a logarithmic connection is

$$d + A(z) \frac{dz}{z}, \quad A(z) = A_0 + \sum_{j \geq 1} A_j z^j, \quad A_0 = \text{diag}(\lambda_i, -\lambda_i).$$

Seek a gauge $G(z) = I + \sum_{j \geq 1} G_j z^j$ taking this connection to the constant-residue model. The equation is

$$zG' = GA_0 - A(z)G,$$

and comparison of coefficients yields

$$(nI + \text{ad}_{A_0})G_n = - \sum_{j=1}^n A_j G_{n-j}, \quad n \geq 1.$$

The eigenvalues of the operator on the left are $n, n, n + 2\lambda_i, n - 2\lambda_i$, all nonzero by ((5.7)). Its inverse has norm at most K/n for one constant K : the bound is immediate for large n , and finitely many smaller n can be absorbed into K . On a sufficiently small parameter neighborhood, the coefficient norms $a_j = \sup \|A_j\|$ have positive convergence radius. If $g_0 = 1$ and

$$ng_n = K \sum_{j=1}^n a_j g_{n-j},$$

then $\|G_n\| \leq g_n$ by induction. The majorant series is

$$\sum_{n \geq 0} g_n z^n = \exp \left(K \sum_{j \geq 1} \frac{a_j}{j} z^j \right),$$

so G converges locally uniformly, holomorphically in the parameters. Since $G(0) = I$, it is invertible after shrinking the disc.

Finally, a single-valued horizontal morphism H between two such constant-residue models satisfies

$$zH' = HA_0 - A_0H.$$

Each matrix coefficient has the form $c z^\beta$, with $\beta = 0$ on the diagonal and $\beta = \pm 2\lambda_i$ off the diagonal. Single-valuedness forces the off-diagonal coefficients to vanish, and the diagonal coefficients are constant. Hence a flat isomorphism extends uniquely across the puncture and preserves the labelled eigenlines. This proves the family correspondence used below in the present nonresonant case.

Every reducible semisimple character is $\nu \oplus \nu^{-1}$. At a puncture,

$$\nu(\gamma_i) \in \{\mu_i, \mu_i^{-1}\}, \quad \prod_{i=1}^n \nu(\gamma_i) = 1. \quad (5.9)$$

There are finitely many admissible sign patterns. For each, the $2g$ handle values of ν are arbitrary elements of $\mathbb{C}^*$.

Pair patterns by simultaneous inversion and choose one representative from each pair. No pattern is its own inverse, because every local pair of eigenvalues is distinct. After this choice there is no remaining involution quotient. Each component is a torus $(\mathbb{C}^*)^{2g}$.

This is an algebraic identification. Choose a puncture loop γ_0 whose selected eigenvalue is μ_0 . For any handle generator a ,

$$\nu(a) = \frac{\text{tr } \rho(\gamma_0 a) - \mu_0^{-1} \text{tr } \rho(a)}{\mu_0 - \mu_0^{-1}}. \quad (5.10)$$

Thus trace coordinates give a regular inverse to the parameterization. The same formula for the other puncture loops recovers their discrete eigenvalue choices; the finitely many resulting components are disjoint and closed.

If $n = 1$, no reducible character exists, because the rank-one relation would force $\nu(\gamma_1) = 1$, contradicting ((5.7)). In that case the whole moduli space is already covered by the irreducible argument.

Suppose a reducible component exists, so $n \geq 2$. The punctured-surface group is free of rank

$$s = 2g + n - 1.$$

Choose one puncture loop γ_0 among free generators; its fixed value satisfies $\nu(\gamma_0)^2 \neq 1$.

Nonsplit extensions with invariant line character ν are classified by nonzero classes in

$$\text{Ext}^1(\nu^{-1}, \nu) = H^1(C \setminus D, \nu^2). \quad (5.11)$$

A cocycle on a free group is determined by arbitrary values at its s generators. Coboundaries change those values by

$$b(\gamma) \mapsto b(\gamma) + (1 - \nu(\gamma)^2)c.$$

Since $1 - \nu(\gamma_0)^2$ is a fixed nonzero scalar, there is a unique choice of c setting $b(\gamma_0) = 0$. The remaining $s - 1$ values are unrestricted.

This gauge gives the following global trivialization over the entire handle-character torus:

$$H^1(C \setminus D, \nu^2) \cong (\mathbb{C}^*)^{2g} \times \mathbb{C}^{2g+n-2}. \quad (5.12)$$

Quotienting nonzero extension classes by the scalar change of the two line factors yields the product with

$$\mathbb{P}^{2g+n-3}.$$

Indeed, after the normalization $b(\gamma_0) = 0$, any further upper-triangular change of frame preserving it has zero unipotent parameter. A diagonal determinant-one change of frame acts on the remaining cocycle coordinates by $b \mapsto t^2 b$. Since $t \mapsto t^2$ surjects onto $\mathbb{C}^*$, the nonzero orbits are precisely scalar projective classes. On the chart where one remaining coordinate b_j is nonzero, use the coordinates b_ℓ/b_j for $\ell \neq j$. These are holomorphic quotient coordinates and have the usual projective transition functions.

The nonresonant logarithmic extension of the rank-one local system ν has residues $\varepsilon_i \lambda_i$, and hence degree

$$\deg L_\nu = - \sum_i \varepsilon_i \lambda_i. \quad (5.13)$$

Choose a nonzero meromorphic section s of L_ν and write $\nabla s = \omega s$. At a point p , if $s = z^{m_p} e$ in a local holomorphic frame, then

$$\omega = m_p \frac{dz}{z} + \frac{\nabla e}{e}.$$

Thus $\text{res}_p \omega = m_p$ away from D , and $\text{res}_{t_i} \omega = m_{t_i} + \varepsilon_i \lambda_i$ at a puncture. Summing residues of the meromorphic one-form ω gives

$$0 = \sum_p m_p + \sum_i \varepsilon_i \lambda_i = \deg L_\nu + \sum_i \varepsilon_i \lambda_i,$$

which proves ((5.13)) with the stated sign convention. The sum is integral by ((5.9)). At each puncture, the induced parabolic weight depends only on the chosen eigenline and the fixed weights, not on the handle monodromy or extension class.

Let $w_i(\nu)$ denote that weight, and let w_i^+, w_i^- be the two weights of the rank-two object. Its stability score is

$$\delta(\nu) = \deg L_\nu + \sum_i w_i(\nu) - \frac{1}{2} \sum_i (w_i^+ + w_i^-).$$

For the opposite line character the degree is the negative and the complementary weights are used, so

$$\delta(\nu^{-1}) = -\delta(\nu). \quad (5.14)$$

Genericity of the weights makes these finitely many scores nonzero.

A nonsplit extension has a unique invariant line. It is stable exactly when the score of that line is negative. Consequently, for each reducible semisimple component exactly one of the two extension directions is stable, and every nonzero extension class in that direction is stable. The split connection has both invariant lines and is unstable because one score is positive.

Thus the stable extension direction is the full projective space in ((5.12)).

Lemma 5.5. *For every component $Z \cong (\mathbb{C}^*)^{2g}$ of the reducible character locus,*

$$\mathrm{RH}^{-1}(Z) \cong (\mathbb{C}^*)^{2g} \times \mathbb{P}^{2g+n-3}$$

as complex manifolds. These inverse images are closed smooth submanifolds of the stable connection moduli space.

Proof. The explicit free-group cocycles in Section 5.2 give holomorphic families of monodromy matrices on affine charts of the projective extension space. Their changes of chart are changes of frame. Applying the family logarithmic correspondence in Section 5.2 gives holomorphic families of parabolic connections, which glue as maps to the coarse stable moduli space.

Conversely, locally choose a frame at a base point for a family of stable reducible connections. The unique invariant line is holomorphic in the family, because the fixed regular semisimple puncture monodromy has distinct eigenlines. Normalize the cocycle at γ_0 as above, and take its remaining projective class. This gives the holomorphic inverse. Therefore the identification is analytic, and for each admissible reducible-character component,

$$\mathrm{RH}^{-1}(\text{that component}) \cong (\mathbb{C}^*)^{2g} \times \mathbb{P}^{2g+n-3}. \quad (5.15)$$

The reducible-character locus is closed, so its inverse image is a closed complex submanifold. $\square$

The reducible-character set is closed in $\mathcal{X}$. Its preimage is, by ((5.15)), a finite disjoint union of smooth Oka manifolds. It is a closed complex submanifold of the smooth connection moduli space. Its complement is the irreducible Oka locus from Section 5.2.

One application of the Oka union theorem [17, Theorem 1.5] proves Theorem 5.4.

6 Elliptic and abelian fibrations

6.1 Elliptic surfaces

Lemma 6.1. *Let C be a reduced curve whose irreducible components are smooth rational curves and whose dual graph is a finite tree; assume each graph edge corresponds to one ordinary intersection point. If a finite group acts on C , then C has a finite invariant closed filtration whose smooth differences are disjoint unions of points, copies of $\mathbb{C}$, and possibly one copy of $\mathbb{P}^1$ per connected tree. If the group acts freely on an Oka manifold A , the diagonal quotient of $C \times A$ has a finite closed filtration with Oka differences.*

Proof. Let Γ be the dual tree and let C_v denote the component indexed by a vertex v . The center of a finite tree is either a vertex or an edge and is preserved by every graph automorphism. If the center is a vertex v_0 , put

$$F_j = \bigcup_{\mathrm{dist}(v, v_0) \leq j} C_v \quad (0 \leq j \leq \mathrm{diam} \Gamma), \quad F_{-1} = \emptyset.$$

Every F_j is a finite union of complete components and hence is closed analytic. For $j \geq 1$, each vertex at distance j has exactly one adjacent vertex at distance $j - 1$; two vertices at distance j cannot be adjacent, since that would create a cycle. Therefore

$$F_0 \cong \mathbb{P}^1, \quad F_j \setminus F_{j-1} \cong \bigsqcup_{\mathrm{dist}(v, v_0)=j} (C_v \setminus \{p_v\}),$$

where p_v is the intersection with the unique preceding component. Each difference is a disjoint union of affine lines. If the center is an edge with endpoints v_0, v_1 and intersection p , first take $F_0 = \{p\}$, then $F_1 = C_{v_0} \cup C_{v_1}$; their difference is two disjoint affine lines. For $j \geq 1$, put

$$F_{j+1} = \bigcup_{\text{dist}(v, \{v_0, v_1\}) \leq j} C_v.$$

The same unique-predecessor calculation applies. Every stage is invariant. Reversing either increasing filtration gives the required descending filtration, and for trees permuted by the group one takes the union of corresponding stages over each orbit.

The product filtration on $C \times A$ descends to the diagonal quotient. Each of its smooth differences is a finite disjoint union of products of A with a point, $\mathbb{C}$, or $\mathbb{P}^1$, and is therefore Oka. The diagonal action is free on every such difference, so its quotient is Oka by permanence under finite unramified coverings. Hence the descended filtration has Oka differences. $\square$

Lemma 6.2. *Let T be a closed discrete subset of $\mathbb{C}^2$ whose projection to the first coordinate is a closed discrete set. Then T is tame in the sense of [48, Definition 3.3].*

Proof. Write the distinct first coordinates as s_j . In the vertical line over s_j , the second coordinates form a closed discrete set T_j . Choose a point $\beta_j \notin T_j$ and a nonzero number a_j such that

$$a_j(T_j - \beta_j) \cap \{|w| \leq j\} = \emptyset.$$

After β_j is chosen, take $|a_j|$ sufficiently large. Avoiding countably many values of a_j also makes this image disjoint from the previously chosen images. Interpolate the values of $\log a_j$ and $-a_j\beta_j$ by entire functions h and b on the first coordinate. The automorphism

$$(s, w) \mapsto (s, e^{h(s)}w + b(s))$$

makes the second-coordinate projection of T injective and closed discrete. If the resulting set is enumerated as (u_k, w_k) , entire interpolation in w gives a shear making the first coordinate equal to k at these points; interpolation in the first coordinate gives another shear making the second coordinate zero. Thus T is carried to a standard discrete subset of an affine line. This vertical–horizontal shear normalization follows the strategy of Rosay–Rudin’s discrete-projection criterion [48, proof of Theorem 3.5]. $\square$

The complement of a tame discrete set in $\mathbb{C}^n$, $n \geq 2$, is Oka, and tame-set avoidance can be accomplished by Fatou–Bieberbach domains; see [20, §6.4].

We now apply these filtration and tameness statements to elliptic surfaces.

Let $f : X \rightarrow B$ be an elliptic fibration on a smooth connected compact complex surface. Arbitrary finite sequences of point blowups, including infinitely near centers, are allowed. Let

$$m_b = \gcd\{\text{multiplicities of the irreducible components of } f^{-1}(b)\},$$

and set

$$\chi_{\text{orb}}(f) = 2 - 2g(B) - \sum_b \left(1 - \frac{1}{m_b}\right).$$

Theorem 6.3.

$$X \text{ is Oka} \iff \chi_{\text{orb}}(f) \geq 0.$$

The criterion applies to the indicated nonminimal surfaces and to their relatively minimal models.

First suppose the fibration is relatively minimal, its base is $\mathbb{C}$, and it has no multiple fibers. Buzzard–Lu’s section theorem gives a holomorphic section. The section may be prescribed to pass through a multiplicity-one component of each singular fiber [13, Lemma 4.4].

Let G^0 be the identity component of the relative Néron model determined by this section; the general theory of Néron models is developed in [10]. It is an open submanifold of X . It includes each entire smooth fiber and the smooth part of the chosen multiplicity-one component in every singular fiber. Its fibers are elliptic curves, $\mathbb{C}^*$, or $\mathbb{C}$. Its Lie algebra is a line bundle on $\mathbb{C}$ and hence holomorphically trivial. Consequently its relative exponential gives

$$\text{Exp} : \mathbb{C}^2 \longrightarrow G^0.$$

The map is surjective and a local biholomorphism everywhere: it is the identity on the base, and its derivative in the fiber direction is the derivative of the exponential of a one-dimensional connected commutative complex Lie group. Thus G^0 is strongly dominable and is Oka by the strong dominability criterion [17, Theorem 1.3].

It remains to fill $X \setminus G^0$. The filtration uses Kodaira’s classification of singular elliptic fibres [38], the reduced support, and the number of punctures on its components.

- For I_1 and II, the omitted set is respectively the node or the cusp.
- For I_r , r at least 2, the omitted components form a rational chain after the chosen component is omitted. The two attachment points on the chosen component remain in the boundary. A leaf-by-leaf filtration has differences $\mathbb{P}^1$ with at most two points removed; all are Oka.
- For III, the other component is a complete $\mathbb{P}^1$ containing the tangency point.
- For IV, the omitted two components meet at the unique triple point. Begin with one complete component and add the other with the intersection point removed.
- For the starred Kodaira types, the reduced support is a rational tree and the chosen multiplicity-one component is a leaf. The boundary is the remaining tree, including its attaching point. Apply Section 6.1, or peel its leaves in reverse order.

These are finite closed analytic filtrations with smooth Oka differences. Therefore the finite Oka stratification theorem [17, Corollary 4.5] proves that X is Oka.

The same proof works when the singular fibers form a locally finite family with only finitely many fiber types and a uniform bound on the number of components. One takes the locally finite union of corresponding stages of the filtrations. Each stratum is a countable disjoint union of Oka components, hence Oka.

For $B = \mathbb{P}^1$ and no multiple fibers, cover B by two affine lines. Buzzard–Lu applies over each affine line, and the preceding proof makes each inverse image Oka. Zariski localization then makes X Oka.

For an elliptic base, pull the fibration back along $\mathbb{C} \rightarrow B$. This is an unramified covering on total spaces. The singular fibers become locally finite copies of the original finite collection, so the preceding argument applies. Oka descent through the covering proves the assertion. The same reasoning treats $\mathbb{C}^*$ bases when noncompact fibrations occur as intermediate spaces.

Place an orbifold point of order m_b at each multiple fiber. For a good nonhyperbolic compact orbifold curve, its universal orbifold cover is $\mathbb{P}^1$ or $\mathbb{C}$. At a cone point the covering has ramification index exactly m_b . Normalizing the fiber product gives an unramified holomorphic covering of total spaces and removes the multiple fibers [14, Proposition 3.4 and proof of Theorem 3.9].

The local unramifiedness follows directly from divisibility of the fiber divisor. Near any point of a smooth total space, write the base coordinate as

$$t = uh^m,$$

where u is a nowhere zero holomorphic function and m is the multiplicity of the fiber. After shrinking, u has an m -th root. The normalization of $s^m = t$ is then locally a disjoint union of copies of the original smooth total space, with s equal to an m -th root of u times h . This remains valid at singular points of the reduced fiber. The normalized cover is therefore unramified throughout the total space. The resulting fiber divisor has component multiplicities divided by m .

The fibration upstairs has no multiple fibers and is Oka by Section 6.1–Section 6.1; the property descends.

The remaining compact orbifold types are $\mathbb{P}^1$ with one cone point, and $\mathbb{P}^1$ with two cone points of unequal orders. Cover $\mathbb{P}^1$ by two affine lines, each containing at most one cone point. If its order is m , use $s \mapsto s^m$ on that affine line and normalize the fiber product. The preceding local calculation gives an unramified total-space covering, and the new base is $\mathbb{C}$ with no multiple fibers. Each of the two inverse images is therefore Oka. Zariski localization then proves the result for these two orbifolds.

If the orbifold Euler characteristic is negative, the orbifold base is uniformized by the unit disc. The same normalized pullback gives an unramified covering $\hat{X} \rightarrow X$ and a nonconstant surjective holomorphic map $\hat{X} \rightarrow \mathbb{D}$.

If X were Oka, then $\hat{X}$ would be Oka. At a point where this map has nonzero differential, strong dominability would give a map $\mathbb{C}^2 \rightarrow \hat{X}$ with surjective differential. Its composite with $\hat{X} \rightarrow \mathbb{D}$ would be a bounded entire function, hence constant, contradicting the chain rule. This proves necessity.

It remains to justify the statement for nonminimal surfaces.

Work first over $\mathbb{C}$. Separate the finite cluster of centers into those lying over the Néron open G^0 and those lying over its complement. Pull the first cluster back along the surjective local biholomorphism $\text{Exp} : \mathbb{C}^2 \rightarrow G^0$. The initial supports form a closed discrete set T contained in a discrete union of vertical fibers. They are tame by Section 6.1. Blowups commute with local biholomorphic base change, so the full finite cluster pulls back to a locally finite cluster of point blowups of $\mathbb{C}^2$ with uniformly bounded depth. Let W be the resulting source.

The complement of the total exceptional divisor in W is $\mathbb{C}^2 \setminus T$ and is Oka. Over each point of T , the total exceptional divisor is a rational tree. Its irreducible components are complete $\mathbb{P}^1$'s, and its depth and number of components are uniformly bounded by those of the original finite cluster. Applying Section 6.1 simultaneously to these locally finite trees gives a finite Oka filtration. Hence W is Oka by the finite Oka stratification theorem [17, Corollary 4.5].

The induced map from W to the modification of G^0 is surjective and locally biholomorphic. For every target point, choose a preimage and a map strongly dominating W there. Composition strongly dominates the target at the desired point. Thus the modified Néron open is Oka by the strong dominability criterion [17, Theorem 1.3].

Centers over $X \setminus G^0$ leave this open set unchanged. Their total transforms add finite rational trees to the boundary configurations already used in Section 6.1. A blowup at a smooth boundary point attaches an exceptional rational curve; a blowup at an intersection subdivides an edge or separates branches; further point blowups repeat these operations. They introduce no new cycle beyond the original Kodaira cycle, whose chosen component was already omitted. The resulting boundary again admits a leaf filtration with at most two punctures on a rational component. In the I_1 and II cases, the omitted singular point is replaced by the corresponding exceptional rational configuration, and the same argument applies. Thus the finite Oka stratification theorem [17, Corollary 4.5] fills the boundary after all blowups.

Under the orbifold covers and the universal covers of elliptic bases, the centers lift to locally finite copies of the original finite cluster. The uniform finite bound still applies. Necessity is unchanged because point blowups preserve the gcd multiplicity of each fiber. This completes the proof of Theorem 6.3.

6.2 Abelian fibrations

The defect function and δ -regularity condition come from Ngô's theory of weak abelian fibrations [43, §7]. We use the curve-base classification of Yoon-Joo Kim [37], which is a 2026 preprint. Let B be a smooth connected complex algebraic curve, and let

$$f : X \longrightarrow B$$

be a flat projective abelian fibration whose generic fiber is a torsor under an n -dimensional abelian variety, with $n \geq 2$. We use the following conditions.

The fibration is **minimal** if X has terminal singularities and K_X is numerically trivial relative to f : its intersection with every proper curve contracted by f is zero. Let $P \rightarrow B$ be the Néron model of the generic fiber's translation group. For its fiberwise identity component, Chevalley's structure theorem writes each fiber as an extension of an abelian variety by a connected linear algebraic group [11, §I.2]. The fibration is **δ -regular** if this connected linear part has dimension at most one at every closed point of B . This is the condition $\delta \leq \text{codim}$ on a curve: the generic defect is zero, and every closed point has codimension one. Equivalently, the abelian part of every fiber has dimension at least $n - 1$.

These are [37, Definitions 2.1.1 and 2.2.1]. The regularity conclusion of [37, Proposition 2.2.4] says that X is already smooth under these assumptions.

Theorem 6.4. *Suppose B is Oka and the minimal δ -regular fibration f has a section. Then X is an Oka manifold.*

A holomorphic section suffices, because it is used only to trivialize the analytic Néron torsor. Thus the theorem applies over $\mathbb{P}^1$, an elliptic curve, $\mathbb{A}^1$, or $\mathbb{C}^$. The section-free extension is proved in Section 6.2.*

Write $X^{\text{sm}/f}$ for the smooth locus of the morphism f . A section σ lies in this open set at every base point, since $df \circ d\sigma = \text{id}$. Thus each fiber has a point smooth over the base. By [37, Proposition 2.2.2(2)],

$$X^{\text{sm}/f} \longrightarrow B$$

is a P -torsor. The section identifies it analytically with P .

The identity component $P^0 \rightarrow B$ is Oka by Theorem 1.9. The complement $P \setminus P^0$ consists of finitely many connected components over the finitely many singular values. Each component is a homogeneous manifold, being a torsor under a connected complex algebraic group. This complement is closed and smooth in P . Hence the Oka union theorem [17, Theorem 1.5] proves that P , and therefore $X^{\text{sm}/f}$, is Oka.

The remaining issue is the geometry of the critical locus. The classification below verifies it for the original minimal model, including its nonreduced fibers.

By [37, Theorem 5.1.1], a singular semistable fiber is a cycle of $\mathbb{P}^1$ -bundles over one $(n - 1)$ -dimensional abelian variety A . Neighboring bundles are glued along their sections; a cycle of length one permits self-identification of the two sections of a single bundle. The gluing may involve translations.

The fiber is reduced. Its singular locus consists precisely of the glued sections, each isomorphic to an abelian variety. These sections are pairwise disjoint as components of the singular locus, including the self-glued case. Thus the reduced critical locus contributed by a semistable fiber is a finite disjoint union of smooth Oka manifolds.

For an unstable fiber, [37, Theorem 5.2.1] gives an isomorphism of fiber schemes

$$X_b \cong (C \times A)/G.$$

Here C is one of the Kodaira curves

$$II, III, IV, I_r^*, II^*, III^*, IV^*,$$

with its scheme multiplicities, A is an abelian variety of dimension $n - 1$, and G is a finite abelian group acting diagonally. Its action on A consists of free translations. In particular, the diagonal quotient is étale even when C is singular or nonreduced.

Let $K = \text{Sing}(C)_{\text{red}}$. Smoothness descends and ascends through an étale morphism, so the reduced critical locus in this fiber is exactly

$$\text{Crit}(f)_{\text{red}} \cap X_b = (K \times A)/G.$$

For the types II, III, IV , the set K is the unique singular point. Thus this locus is an abelian quotient A/G , which is smooth and Oka.

For a starred type, K is the reduced union of the components whose multiplicity is at least two. Every intersection point involving a multiplicity-one component lies on this union, whose dual graph is a finite rational tree.

Apply the invariant-center filtration of Section 6.1 to this tree. The graph center is preserved by G , and so are its successive distance shells. A vertex center contributes an entire $\mathbb{P}^1$; an edge center contributes its intersection point followed by the two adjacent affine lines; each later component contributes an affine line. All partial unions are closed and invariant. After taking their products with A , the smooth differences are finite disjoint unions of

$$A, \quad \mathbb{A}^1 \times A, \quad \mathbb{P}^1 \times A.$$

The group acts freely on each stratum because it acts freely on A , so their quotients are Oka. If a finite group permutes several trees, run the filtration simultaneously on each orbit of connected trees; its invariance and closedness are unchanged.

We have therefore obtained a finite closed Oka filtration of the critical locus in every unstable fiber. Combining the finitely many bad fibers and adjoining $X^{\text{sm}/f}$ as the open stratum gives a finite closed Oka filtration of X . Theorem 6.4 follows from the finite Oka stratification theorem [17, Corollary 4.5].

We next remove the assumption that the abelian fibration has a global section.

All sheaves and sheaf cohomology in this section are taken on the classical analytic topology. The following lemma treats the period sheaves occurring at singular fibres.

Lemma 6.5. *Let $D \subset \mathbb{C}$ be closed discrete, possibly infinite. Let $\mathcal{F}$ be a sheaf of abelian groups whose restriction to $U = \mathbb{C} \setminus D$ is a local system $\mathcal{L}$. Then*

$$H^2(\mathbb{C}, \mathcal{F}) = 0.$$

Proof. Let $j : U \hookrightarrow \mathbb{C}$. The kernel and cokernel of the natural map $\mathcal{F} \rightarrow j_*\mathcal{L}$ are supported on D . A sheaf supported on a closed discrete subset is the pushforward of a sheaf on that discrete space; its positive-degree cohomology is zero. Applying the two short exact sequences through the image of this map gives

$$H^2(\mathbb{C}, \mathcal{F}) \cong H^2(\mathbb{C}, j_*\mathcal{L}).$$

The relevant Leray exact sequence is

$$H^1(U, \mathcal{L}) \longrightarrow H^0(\mathbb{C}, R^1j_*\mathcal{L}) \longrightarrow H^2(\mathbb{C}, j_*\mathcal{L}) \longrightarrow H^2(U, \mathcal{L}) = 0.$$

The last equality holds because a punctured plane has the homotopy type of a one-dimensional CW complex. Choose paths to small circles around each $d \in D$, identify the local-system fibers with one module V , and denote the corresponding monodromies by T_d . Then

$$H^0(\mathbb{C}, R^1 j_* \mathcal{L}) = \prod_{d \in D} V / (T_d - 1)V.$$

The fundamental group of $\mathbb{C} \setminus D$ is free with a basis of the puncture loops γ_d . This remains true for an infinite closed discrete set: use an exhaustion by disks whose boundaries avoid D , extending the finite free basis whenever new punctures enter. For a prescribed tuple of coinvariant classes, choose representatives $v_d \in V$. Define a crossed homomorphism by

$$c(\gamma_d) = v_d, \quad c(\gamma_d^{-1}) = -T_d^{-1}v_d, \quad c(gh) = c(g) + \rho(g)c(h),$$

where ρ is the monodromy representation. Equivalently, for a reduced word $w = s_1 \cdots s_\ell$,

$$c(w) = \sum_{i=1}^{\ell} \rho(s_1 \cdots s_{i-1})c(s_i).$$

Insertion or deletion of $\gamma_d \gamma_d^{-1}$ changes this sum by $\rho(g)(v_d + T_d(-T_d^{-1}v_d)) = 0$. Thus the formula is well-defined and satisfies the crossed-homomorphism identity. Every word is finite, even when D is infinite, so no convergence requirement is imposed on the tuple (v_d) . The corresponding class in $H^1(U, \mathcal{L})$ restricts at d to $v_d \bmod (T_d - 1)V$. It therefore realizes the prescribed tuple.

The first arrow in the Leray segment is therefore surjective, which proves the lemma. $\square$

Let $P \rightarrow \mathbb{C}$ be a smooth commutative Néron group family arising from the preceding fibrations or their analytic covering pullbacks, and let P^0 be its identity component. Its generic fibres are abelian varieties, and its singular values form a closed discrete set D .

Proposition 6.6. *Every analytically locally trivial P -torsor over $\mathbb{C}$ has a holomorphic section.*

Proof. Write $\mathcal{P}^0$ for its sheaf of holomorphic sections, and let $\mathcal{V}$ be the sheaf of holomorphic sections of $\text{Lie}(P/\mathbb{C})$. The relative exponential gives an exact sequence of sheaves

$$0 \longrightarrow \Lambda \longrightarrow \mathcal{V} \xrightarrow{\exp} \mathcal{P}^0 \longrightarrow 0.$$

Surjectivity here is local on the base. The exponential of each connected commutative complex Lie group is surjective. At a chosen logarithm its derivative is an isomorphism in the fiber directions, so the holomorphic inverse function theorem produces a local logarithm of a nearby holomorphic section. This proves the stated sheaf exactness even when the toric or unipotent rank changes.

Away from D , Λ is the integral period local system. Lemma 6.5 gives $H^2(\mathbb{C}, \Lambda) = 0$. Cartan's theorem B [20, §2.6] gives $H^1(\mathbb{C}, \mathcal{V}) = 0$, since $\mathcal{V}$ is locally free coherent. The long exact sequence therefore yields

$$H^1(\mathbb{C}, \mathcal{P}^0) = 0.$$

The component-group sheaf $\mathcal{P}/\mathcal{P}^0$ is supported on the closed discrete set D and has no positive-degree cohomology. The component map is locally surjective on holomorphic sections, since the smooth group family has local sections through every point. A second long exact sequence gives

$$H^1(\mathbb{C}, \mathcal{P}) = 0.$$

Analytically locally trivial P -torsors are classified by $H^1(\mathbb{C}, \mathcal{P})$, where $\mathcal{P}$ is the sheaf of holomorphic sections. Hence every such torsor has a holomorphic section. $\square$

Proposition 6.7. *Let $f : X \rightarrow B$ be a minimal δ -regular abelian fibration over an Oka algebraic curve, without a section assumed. If*

$$m_b = \gcd\{\text{multiplicities of the prime components of } f^{-1}(b)\} = 1$$

for every $b \in B$, then X is Oka.

Proof. Kim [37, Lemma 2.3.3] proves that a fibre of multiplicity one has a reduced component. Hence $m_b = 1$ gives a multiplicity-one component; a general point of that component is smooth over the base. If all $m_b = 1$, [37, Proposition 2.2.2(2)] identifies $X^{\text{sm}/f}$ with a P -torsor. Over $B = \mathbb{C}$, Section 6.2 supplies a holomorphic section, so the proof of Section 6.2 applies.

For $B = \mathbb{C}^*$ or an elliptic curve, pull back to its universal covering $\mathbb{C}$. The same torsor argument applies there, and the Oka conclusion descends through the unramified total-space covering. The preimage of the discriminant is a closed discrete set, so Lemma 6.5 applies.

The boundary still has a finite filtration: the pullback contains only copies of the finitely many original bad-fiber types. Choose a finite filtration for each type and use its locally finite disjoint union over all corresponding lifted values. There are finitely many types and finitely many stages for each. Run the filtrations of the disjoint fibre types successively, separating strata of different dimensions. Each stratum is a countable disjoint union of Oka components, hence is Oka.

For $B = \mathbb{P}^1$, remove one smooth fiber F_∞ . The remaining base is $\mathbb{C}$, so $X \setminus F_\infty$ is Oka. The Oka union theorem [17, Theorem 1.5] fills the fibre back. □

Now let B be compact and allow arbitrary multiplicities m_b . The exact base-change result needed is [37, Lemma 7.1.2]: after a base change totally ramified to degree m_b at a fiber of multiplicity m_b , the normalized total-space pullback is étale over X , remains minimal and δ -regular, and has nonmultiple fiber there. The preservation of the defect condition uses the base-change results of [37, Corollary 6.1.5].

The étale assertion can also be seen directly, and is useful for both finite and infinite orbifold covers. Since X is smooth and the fiber divisor is $m_b D_b$ for an integral Cartier divisor D_b , locally

$$t = u h^{m_b},$$

where t is a parameter at b , h defines D_b , and u is a holomorphic unit. On a sufficiently small neighborhood choose an m_b -th root of u . The equation $s^{m_b} = t$ then factors into m_b distinct branches

$$s = \zeta u^{1/m_b} h.$$

Normalization separates these branches, and each is a copy of the original neighborhood in X . Hence the projection from the normalization is a local biholomorphism even over singular points of the reduced fiber support. In the new base parameter, the fiber divisor has component multiplicities divided by m_b , so their greatest common divisor is one.

Minimality is preserved because canonical bundles pull back through étale maps. Kim's classification and base-change theorem preserve δ -regularity. These statements are local and therefore apply to the normalized pullback along an analytic orbifold covering as well.

Define

$$\chi_{\text{orb}}(f) = 2 - 2g(B) - \sum_b \left(1 - \frac{1}{m_b}\right).$$

Theorem 6.8. *Let $f : X \rightarrow B$ be a minimal δ -regular abelian fibration of relative dimension $n \geq 2$ over a compact complex algebraic curve. Then*

$$X \text{ is Oka} \iff \chi_{\text{orb}}(f) \geq 0.$$

Proof. Sufficiency for good orbifolds. A compact good orbifold curve of nonnegative orbifold Euler characteristic has universal orbifold covering $\mathbb{P}^1$ or $\mathbb{C}$. Its ramification at each marked point is exactly the prescribed order m_b . Normalize the pullback of X to this covering. By Section 6.2, the projection to X is unramified, and the lifted fibration is minimal, δ -regular, and nonmultiple. Its total space is Oka by Section 6.2, including the analytic $\mathbb{C}$ -cover case and its uniform finite boundary filtration. Oka descent proves that X is Oka.

Bad orbifolds. The compact bad orbifolds are $\mathbb{P}^1$ with one cone point, or with two cone points of unequal orders. They have positive orbifold Euler characteristic but lack the universal covering just used. Cover $\mathbb{P}^1$ by two algebraic affine lines such that each contains at most one cone point. Over one such line, put its possible cone point at 0, with order m , and use the map

$$\mathbb{C} \longrightarrow \mathbb{C}, \quad s \longmapsto s^m.$$

If there is no cone point, take $m = 1$. The normalized total-space pullback is unramified over the original open total space, minimal, δ -regular, and nonmultiple. Its base is $\mathbb{C}$, so Section 6.2 makes it Oka. Descent makes the original open total space Oka. The two opens are Zariski open in X and cover X ; Oka localization proves the result.

The same construction handles two cone points of equal orders directly.

Necessity. If $\chi_{\text{orb}}(f) < 0$, the compact orbifold is good and its universal orbifold covering is the disk. Normalizing the total-space pullback gives an unramified covering $\tilde{X} \rightarrow X$ together with a nonconstant holomorphic map $\tilde{f} : \tilde{X} \rightarrow \mathbb{D}$. If X were Oka, then $\tilde{X}$ would be Oka and hence strongly dominable by the strong dominability criterion [17, Theorem 1.3]. Choose a point where $d\tilde{f} \neq 0$, and a map from $\mathbb{C}^{n+1}$ strongly dominating $\tilde{X}$ there. Its composition with $\tilde{f}$ is a bounded entire function, hence constant, contradicting the surjectivity of the dominating differential. Thus X is not Oka. $\square$

6.3 The Gross–Popescu Calabi–Yau pair

For a sufficiently general parameter $y \in \mathbb{P}^2$, Gross and Popescu [30, §6] construct a complete intersection $V_{8,y} \subset \mathbb{P}^7$ of four quadrics with 64 ordinary double points and a smooth projective small resolution

$$X = V_{8,y}^1 \longrightarrow V_{8,y}.$$

The threefold X is Calabi–Yau and carries a morphism $\pi : X \rightarrow \mathbb{P}^1$ whose general fibre is a $(1, 8)$ -polarized abelian surface. Its eight singular fibres are elliptic translation scrolls with smooth elliptic singular loci $E_1, \dots, E_8$. A section gives

$$G = X \setminus (E_1 \sqcup \dots \sqcup E_8)$$

the structure of a smooth group scheme over $\mathbb{P}^1$ with connected fibres. A fibrewise translation action of $G_8 = (\mathbb{Z}/8\mathbb{Z})^2$ on X is free. These facts are proved in [31, Theorem 1.1, Lemma 1.2, Proposition 1.3, and Theorem 1.4]; Schnell identifies the quotient $Y = X/G_8$ with the relative dual construction and proves $D^b(X) \simeq D^b(Y)$ [49, Theorem 4.1 and Lemmas 5.1 and 5.4].

Theorem 6.9. *The Gross–Popescu threefolds X and $Y = X/G_8$ are smooth projective Oka Calabi–Yau threefolds.*

Proof. The group family $G \rightarrow \mathbb{P}^1$ satisfies Theorem 1.9, so G is Oka. Each E_i is an elliptic curve and hence Oka. Their disjoint union

$$E = \bigsqcup_{i=1}^8 E_i$$

is a closed Oka submanifold of X with complement G . The Oka union theorem [17, Theorem 1.5] gives that X is Oka.

The free action makes $X \rightarrow Y$ a finite unramified covering, so Y is Oka. The action preserves the relative volume form and fixes the base; the holomorphic three-form on X therefore descends to a nowhere-vanishing holomorphic three-form on Y . Projectivity and the remaining Calabi–Yau properties follow from the Gross–Popescu construction. $\square$

7 Compactified Jacobians

Compactifications of generalized Jacobians by rank-one torsion-free sheaves were developed by D’Souza [16] and Oda–Seshadri [44]. The relative compactified-Picard and Abel constructions used here are due to Altman–Kleiman [1].

Theorem 7.1. *Let $f : S \rightarrow \mathbb{P}^1$ be a flat projective morphism from a smooth projective complex surface, with a section through the relative smooth locus. Suppose every fibre is geometrically integral of arithmetic genus $g \geq 1$, and every singular fibre has exactly one ordinary node, with local equation $xy = t$. Then the relative compactified Jacobian*

$$\pi : \overline{J}^d(S/\mathbb{P}^1) \longrightarrow \mathbb{P}^1$$

of rank-one torsion-free sheaves of degree d is a smooth projective Oka manifold, for every integer d . For every $n \geq 2g - 1$, the relative Hilbert scheme $\text{Hilb}^n(S/\mathbb{P}^1)$ is also smooth, projective, and Oka.

Altman–Kleiman’s compactified-Picard theorem [1, Theorems (2)–(4)] supplies the projective relative moduli space and its Abel construction. For the one-node local description, see Smith [53, §3.2, Theorem 3.4 and Proposition 3.8].

If C is a one-node fibre with normalization $\nu : \tilde{C} \rightarrow C$ and preimages p, q of the node, its degree- d invertible sheaves form a $\mathbb{C}^*$ -bundle over $\text{Pic}^d(\tilde{C})$. The compactification is obtained by extending to a $\mathbb{P}^1$ -bundle and identifying the zero and infinity sections by translation by $\mathcal{O}(p - q)$. Every non-invertible rank-one torsion-free sheaf is uniquely $\nu_* L$ for $L \in \text{Pic}^{d-1}(\tilde{C})$. The degree shift follows from

$$\chi(\nu_* L) = \chi(L), \quad \chi(\mathcal{O}_C) = \chi(\mathcal{O}_{\tilde{C}}) - 1.$$

The relative deformation at such a sheaf has the analytic model

$$\{uv = t\} \times \mathbb{C}^{g-1}.$$

Eliminating t proves that the total space is smooth. The non-invertible locus is $u = v = 0$, of codimension two in that total space.

Proof of the Oka assertion.

First take $d = 0$. Let

$$G = \text{Pic}^0(S/\mathbb{P}^1).$$

It is the invertible-sheaf locus, including the generalized Jacobians of the singular fibres. It is a smooth connected commutative complex Lie-group family over the entire base, and

$$\text{Lie}(G) = R^1 f_* \mathcal{O}_S.$$

Section 1.3 proves that G is Oka. Equivalently, the Lie bundle has Oka total space, and its relative exponential is a surjective holomorphic submersion: for a connected commutative complex Lie group the exponential is its universal covering homomorphism. Regular-preimage descent gives the same conclusion.

The complement $\bar{J}^0 \setminus G$ is a finite disjoint union of the compact complex tori $\text{Pic}^{-1}(\tilde{C}_b)$, of dimension $g - 1$. The local model above proves that this union is a closed smooth submanifold of the smooth total space. Each component is Oka. Therefore the Oka union theorem [17, Theorem 1.5] gives $\bar{J}^0$ Oka.

The section of f is a Cartier divisor on S and has degree one on every fibre. Tensoring by its d -th power identifies $\bar{J}^0$ with $\bar{J}^d$. This proves all degrees.

Proof of the large-degree Hilbert assertion, including the boundary $n = 2g - 1$.

Let C be any fibre and F a rank-one torsion-free sheaf of degree $n \geq 2g - 1$. Since C is integral and Gorenstein, Serre duality gives

$$H^1(C, F)^* = \text{Hom}(F, \omega_C).$$

A nonzero homomorphism on the right is generically injective. Its kernel is torsion and hence zero; its cokernel has finite length. It would imply $\deg F \leq \deg \omega_C = 2g - 2$, a contradiction. Thus

$$H^1(C, F) = 0, \quad h^0(C, F) = n - g + 1.$$

A nonzero section $s : \mathcal{O}_C \rightarrow F$ is injective, again by integrality and torsion-freeness. Its cokernel T has length n . On a Gorenstein curve, torsion-free sheaves are maximal Cohen–Macaulay, their double dual is themselves, and their local $\text{Ext}^1(-, \mathcal{O}_C)$ vanishes. Dualizing gives

$$0 \rightarrow F^\vee \rightarrow \mathcal{O}_C \rightarrow \mathcal{E}xt^1(T, \mathcal{O}_C) \rightarrow 0.$$

The last sheaf has length n by local Gorenstein duality. Hence $F^\vee$ is the ideal of a unique length- n subscheme. Conversely, dualizing the ideal of a length- n subscheme gives a rank-one torsion-free degree- n sheaf with a nonzero section. The two operations are inverse, modulo multiplication of the section by a nonzero scalar.

After rigidifying along the given section, a universal sheaf exists; alternatively, the same argument works in local universal charts and the projectivizations glue. Vanishing of H^1 and constancy of h^0 imply by cohomology and base change [4, Tag 07VJ] that the section spaces form a vector bundle of rank $n - g + 1$. Thus the Abel morphism is a holomorphically locally trivial $\mathbb{P}^{n-g}$ -bundle over $\bar{J}^n$. It is smooth and projective, and the Oka bundle theorem applies. $\square$

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